Honors Calculus

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Contents

1 Need for calculus **2**
1.1 Making sense of numbers . . . 2
1.2 Making sense of "infinite sums" . . . 3

2 Numbers of all sorts **4**
2.1 Natural numbers . . . 4
2.2 Field . . . 6
2.3 Inequalities . . . 7
2.4 Mathematical Induction . . . 10

3 Sequence and limits **11**
3.1 Decimal expansion . . . 11
3.2 Discussing limits rigorously . . . 15
3.2.1 Back to our scheduled program on the Dichotomy Paradox 17
3.3 Algebraic limit theorem . . . 19
3.4 Cauchy sequence . . . 20
3.5 Bolzano-Weistrass . . . 23

4 Oral presentation: A tussle between logic and intuition in mathematics **27**
4.1 Propositional Calculus . . . 28
4.1.1 Equivalence . . . 28
4.1.2 Negation . . . 29
4.1.3 Conjunction and Disjunction . . . 30
4.1.4 Implication . . . 30

5 Functions **31**
5.1 Graphs . . . 31

6 Limits **33**

7 Continuous functions **42**
7.1 The Least Upper Bound Property of the Reals 43
7.2 Intermediate value theorem using least upper bounds 46
7.3 Uniform continuity 49

8 Derivatives and integrals **51**
8.1 Differentials of some common functions 52
8.2 Critical points, local maxima-minimal and the Mean value theorem 60

9 Inverse functions **69**
9.0.1 Problem set 2 71

10 Integrals **72**

11 Fundamental theorem of Calculus **83**

12 Trignometric functions **85**
12.1 What does angle mean? 86

13 Taylor series **91**
13.0.1 Taylor Polynomials and Taylor's Theorem 92
13.0.2 Pointwise Convergence Does Not Preserve Continuity . . 95
13.1 Problem Set 3 100

These notes are meant to be complementary to the textbook Single Variable Calculus by Michael Spivak [Spi06]. We will also occasionally refer to [Abb16]. These are not complete and students are strongly encouraged to also read the main textbook. Concepts or problems not present in the textbook will be explained in detail in these notes.

1 Need for calculus

1.1 Making sense of numbers

Abstract mathematics requires a proper understanding of every single concept or word. For instance, we have heard of the number π and the number e. We have often heard that π is approximately $\frac{22}{7}$ or 3.14, but that is just an approximation, what does it really mean? You might have heard that π is ratio of length of circumference of a circle divided by the length of the diameter of the circle. But, what does *length* mean? What does *division* mean? There are so many possible circles, which circle do we take? These are all very natural questions that mathematicians from different centuries have tried to understand and answer. The understanding of Calculus starts with these questions. We all have an intuitive understanding of numbers, in this course, we will learn to convert that understanding into precise mathematical statements.

1.2 Making sense of "infinite sums"

Zeno's paradox: Imagine you had to run a distance of L meters. Zeno argues first that to cover this length L, you must first cover half the length $\frac{L}{2}$. At the instant you reach this point, there is exactly $L - \frac{L}{2} = \frac{L}{2}$ remaining of the path: and in order to clear this amount, you must first cover half of the remaining amount $\frac{L}{2}/2 = \frac{L}{4}$. Applying the argument once again at the point of covering $\frac{L}{2} + \frac{L}{4}$ of the remaining distance, we see that you must again cover at least $\frac{L}{8}$ more. Iterating this process, you see Atalanta must hit an infinite number of points at distances $p_k = \frac{L}{2} + ... + \frac{L}{2^k} < 1$ before you cross the entire path. Zeno now argues that this motion must have *at least* entailed performing an infinite number of tasks: and clearly, an infinite number of tasks must take an infinite time to perform. He concludes that the concept of "motion" itself is impossible.

Some Strange Infinite Series

The usual definition of an infinite series is that

$$a_1 + a_2 + a_3 + \cdots$$

is the limit of its partial sums, provided that limit exists. Sometimes the partial sums do not converge. Nevertheless, mathematicians have invented various ways of assigning values to certain divergent series.

Example 1. Consider the series

$$1 - 1 + 1 - 1 + 1 - 1 + \cdots .$$

Assuming that rules of infinite arithmetic are the same as finite arithmetic, we can get various answers:

$$1 - 1 + 1 - 1 + 1 - 1 + \cdots = (1 - 1) + (1 - 1) + \ldots = 0 + 0 + 0 + \ldots = 0.$$

or

$$1 - 1 + 1 - 1 + 1 - 1 + \cdots = 1 + (-1 + 1) + (-1 + 1) \ldots = 1.$$

Different viewpoints lead to different answers:

Cesàro sum: The average of the partial sums approaches

$$\frac{1}{2},$$

so the Cesàro sum is

$$1 - 1 + 1 - 1 + \cdots = \frac{1}{2}.$$

Different methods give different answers. This tells us that there is something weird going on about this infinite sum. Perhaps the usual rules of arithmetic do not apply here.

Problem 1. *Use resources online to argue for different possible values of* $1 + 2 + 3 + \ldots$ *assuming that standard laws of finite arithmetic hold true for infinite arithmetic. One of the answers is also often called Ramanujan's sum.*

There is definitely something very different about Zeno's example from Grandi's series and Ramanujan's sum. Can you point out some differences?

2 Numbers of all sorts

2.1 Natural numbers

We want to build everything from scratch. Including the notion of natural numbers and addition. We won't attempt to show that they exist, rather give a theoretical framework by describing a set of axioms that would define natural numbers. We would later define the notion of addition and multiplication on them.

Definition 1 (Peano Axioms)**.** *The set of natural numbers* $\mathbb{N}$ *is a set together with a distinguished element* $0 \in \mathbb{N}$ *and a function*

$$S : \mathbb{N} \to \mathbb{N},$$

called the successor function*, satisfying the following axioms:*

1. 0 *is a natural number.*
2. *For every* $n \in \mathbb{N}$*, the successor* $S(n)$ *is a natural number.*
3. *There is no natural number whose successor is* 0*; that is,*

$$S(n) \neq 0 \qquad \text{for all } n \in \mathbb{N}.$$

4. *Distinct natural numbers have distinct successors; that is,*

$$S(m) = S(n) \implies m = n.$$

5. *(Principle of Mathematical Induction) If* $A \subseteq \mathbb{N}$ *satisfies*
   (a) $0 \in A$,
   (b) *whenever* $n \in A$*, we also have* $S(n) \in A$,

   then $A = \mathbb{N}$.

Define $1 := S(0), 2 := S(S(0)), 3 := S(S(S(0)))$

Problem 2. *Why can't* $3 = 0$*? Can you state the issue using the Peano axioms?*

Definition 2 (Recursive Definition)**.** *A* recursive definition *is a definition in which elements of a set are defined in terms of previously defined elements. Such a definition is built up step-by-step rather than being given all at once.*

Let us look at some examples.

Definition 3 (Addition)**.** *We define the operation of addition, denoted by* $+$*, recursively as follows.*

For every $m \in \mathbb{N}$*, we set*

$$0 + m = m.$$

Suppose that $n + m$ *has already been defined for* $n, m \in \mathbb{N}$*. Then we define*

$$S(n) + m = S(n + m).$$

Problem 3. *Using the above recursion, is the notion of* $1 + m$ *defined? How about* $2 + m$*?*

Problem 4. *Define* $1.m = m$ *and* $0.m = 0$*. Can you use the notion of "addition" and recursion to define* $m.n$ *for any numbers* m, n*?*

Definition 4 (Principle of Mathematical Induction)**.** *Let* $P(n)$ *be a statement depending on a natural number* n*. Suppose that*

1. $P(0)$ *is true;*
2. *for every* $n \in \mathbb{N}$*, if* $P(n)$ *is true, then* $P(n + 1)$ *is also true.*

Then $P(n)$ *is true for every* $n \in \mathbb{N}$*.*

Proofs by induction are usually organized into the following steps.

1. Identify the statement $P(n)$ that you wish to prove for all $n \in \mathbb{N}$.
2. **Base Case:** Verify that $P(0)$ is true.
3. **Inductive Step:** Assume that $P(n)$ is true for some arbitrary $n \in \mathbb{N}$. This assumption is called the *induction hypothesis*. Using this assumption, prove that $P(n + 1)$ is also true.

Once these three steps have been completed, the principle of mathematical induction guarantees that $P(n)$ is true for every natural number n.

Problem 5. *Show that* $n \neq S(S(S(n)))$ *for a natural number* n*.*

Problem 6. *(Proving associativity) Using principle of mathematical induction prove that*

$$(a + b) + c = a + (b + c)$$

for any natural numbers a, b, c*.*

2.2 Field

A good way to have a formal definition of numbers is to define the properties of numbers. These properties come from our intuitive understanding of what numbers mean. We first define a noion of a field.

First of all, a field should be a set S consisting of elements. If a is an element of S, we write $a \in S$. S is a called a field if it has the operations:

- The set S is equipped with a notion of addition $+$. For $a, b \in S$, there is a definition of $a + b$ such that $a + b \in S$
- The set S is equipped with a notion of addition .. For $a, b \in S$, there is a definition of $a.b$ such that $a.b \in S$

that satisfy the following properties:

1. (Associativity) If a, b, and c are any elements of S, then

   $$a + (b + c) = (a + b) + c.$$

   Problem 7. *From our usual intuition of what numbers should mean, we should also have $a + (b + c) + d = (a + b) + (c + d)$. Do we need to list this property out separately or should it follow from Property 1 above?*

2. There is an element $0 \in S$ that satisfies the following property:

   $$a + 0 = 0 + a = a$$

   for every $a \in S$

   Problem 8. *The above property says that there exists a 0. But a priori one can have two different zeroes, $0_1, 0_2$. Show that $0_1 = 0_2$ using property 2.*

3. For every element $a \in S$, there exists a element $-a \in S$ such that

   $$(-a) + a = 0$$

   Problem 9. *Show that there exists a unique negative of a element a.*

4. (Commutativity) If $a, b \in S$, then

   $$a + b = b + a$$

5. If a, b, and c are any elements, then

   $$a \cdot (b \cdot c) = (a \cdot b) \cdot c.$$

6. If a is any element, then

$$a \cdot 1 = 1 \cdot a = a.$$

Moreover,

$$1 \neq 0.$$

7. For every element $a \neq 0$, there is a element a^{-1} such that

$$a \cdot a^{-1} = a^{-1} \cdot a = 1.$$

8. (Distributivity) If a and b are any elements, then

$$a \cdot b = b \cdot a.$$

9. If a, b, and c are any elements, then

$$a \cdot (b + c) = a \cdot b + a \cdot c.$$

Notice that the equation

$$(b + c) \cdot a = b \cdot a + c \cdot a$$

is also true, by P8.

Problem 10. *Show that* $a.0 = 0$

Problem 11. *Show that if* $a.b = 0$*, then either* $a = 0$ *or* $b = 0$

Problem 12. *Does the set* $\mathbb{Z} := \{\ldots, -2, -1, 0, 1, 2, 3, \ldots\}$ *with the usual notion of addition and multiplication form a field Why or why not?*

Definition 5. *A number* x *is called a rational number if* $x = \dfrac{a}{b}$*, where* a, b *are integers.*

Problem 13. *Does the set of rational numbers denoted by* $\mathbb{Q}$ *form a field?*

2.3 Inequalities

Definition 6. *What does an inequality mean? Talking about what something like* $a > b$ *could mean amounts to seeing whether* $a - b$ *is positive or negative. Thus, giving a notion of inequality for a field* F *amounts to defining a set of positive numbers. A set* P *is called a collection of positive numbers if the following properties are satisfied:*

(P10) (Trichotomy law) For every number a*, one and only one of the following holds:*

(i) $a = 0$*;*

(ii) a is in the collection P;

(iii) $-a$ is in the collection P.

(P11) (Closure under addition) If a and b are in P, then $a+b$ is in P.

(P12) (Closure under multiplication) If a and b are in P, then ab is in P.

These three properties should be complemented with the following definitions:

$$\begin{aligned} a > b &\quad \text{if } a - b \in P; \\ a < b &\quad \text{if } b > a; \\ a \geq b &\quad \text{if } a > b \text{ or } a = b; \\ a \leq b &\quad \text{if } a < b \text{ or } a = b. \end{aligned}$$

Note, in particular, that $a > 0$ if and only if $a \in P$.

All the familiar facts about inequalities, however elementary they may seem, are consequences of (P10)–(P12). For example, if a and b are any two numbers, then precisely one of the following holds:

(i) $a - b = 0$;

(ii) $a - b$ is in the collection P;

(iii) $-(a-b) = b - a$ is in the collection P.

Using the definitions just made, it follows that precisely one of the following holds:

(i) $a = b$;

(ii) $a > b$;

(iii) $b > a$.

The fact that $-a > 0$ if $a < 0$ motivates the following definition. For any real number a, its *absolute value*, denoted by $|a|$, is defined by

$$|a| = \begin{cases} a, & \text{if } a \geq 0, \\ -a, & \text{if } a < 0. \end{cases}$$

For example,

$$|5| = 5, \qquad |-8| = -(-8) = 8, \qquad \left|-\frac{3}{4}\right| = \frac{3}{4}.$$

Theorem 1 (Triangle Inequality)**.** *For all real numbers a and b, we have*

$$|a+b| \leq |a| + |b|.$$

Proof. We consider four cases.

If $a \geq 0$ and $b \geq 0$, then $a + b \geq 0$, so

$$|a + b| = a + b = |a| + |b|.$$

If $a \leq 0$ and $b \leq 0$, then $a + b \leq 0$, so

$$|a + b| = -(a + b) = -a - b = |a| + |b|.$$

Suppose that $a \geq 0$ and $b \leq 0$. Then $|a| + |b| = a - b$. If $a + b \geq 0$, then

$$|a + b| = a + b \leq a - b,$$

because $b \leq 0$. If $a + b \leq 0$, then

$$|a + b| = -(a + b) = -a - b \leq a - b,$$

because $a \geq 0$. Thus, in either case,

$$|a + b| \leq |a| + |b|.$$

Finally, if $a \leq 0$ and $b \geq 0$, the same argument applies after interchanging a and b. □

Problem 14. *Let S be the set of rational numbers $\mathbb{Q}$. Find all numbers that satisfy $x + 3^x < 4$.*

Problem 15. *(Unintuitive example 1) Suppose that we interpret "number" to mean either 0 or 1, and $+$ and $\cdot$ to be the operations defined by the following two tables:*

$+$	0	1
0	0	1
1	1	0

$\cdot$	0	1
0	0	0
1	0	1

Check that properties P1–P9 all hold, even though

$$1 + 1 = 0.$$

to verify that $S = \{0, 1\}$ forms number. Does it make sense to use the inequality $0 < 1$ here? (Think about properties of inequality).

Problem 16 (Unintuitive example 2)**.** *Imagine a 12-hour clock. If the current time is 1 o'clock, then 14 hours later the clock reads 3 o'clock, not 15 o'clock, because*

$$1 + 14 = 15 = 12 + 3.$$

Similarly, if the current time is 4 o'clock, then 33 hours later the clock reads 1 o'clock, because

$$4 + 33 = 37 = 3 \cdot 12 + 1.$$

In other words, we keep only the remainder after division by 12.

Consider the set

$$S = \{0, 1, 2, \ldots, 11\}.$$

For $a, b \in S$*, define*

$$a + b = \textit{the remainder of the usual sum } a + b \textit{ upon division by } 12,$$

and

$$ab = \textit{the remainder of the usual product } ab \textit{ upon division by } 12.$$

For example,

$$8 + 7 = 3, \qquad 5 \cdot 7 = 11$$

in this system.

Do properties P1–P9 still hold for these operations? If not, which property fails?

Problem 17. *The maximum of two numbers* x *and* y *is denoted by* $\max(x, y)$*. For example,*

$$\max(-1, 3) = \max(3, 3) = 3$$

and

$$\max(-1, -4) = \max(-4, -1) = -1.$$

The minimum of x *and* y *is denoted by* $\min(x, y)$*. Prove that*

$$\max(x, y) = \frac{x + y + |x - y|}{2}, \qquad \min(x, y) = \frac{x + y - |x - y|}{2}.$$

Problem 18. *Give a short proof:*

$$|x - y| \leq |x| + |y|$$

Problem 19. *Prove that, for all real numbers* x *and* y*,*

$$\big||x| - |y|\big| \leq |x - y|.$$

Hint: A very short proof is possible if you write things in the right way.

2.4 Mathematical Induction

Mathematical induction is a method for proving that a statement $P(n)$ is true for every integer $n \geq n_0$.

To prove $P(n)$ by induction, we must complete two steps:

1. **Base case.** Prove that $P(n_0)$ is true.
2. **Inductive step.** Assume that $P(k)$ is true for some arbitrary integer $k \geq n_0$. Using this assumption, prove that $P(k + 1)$ is true.

The assumption that $P(k)$ is true is called the *inductive hypothesis*. It is only an assumption within the inductive step; the base case and inductive step together show that the statement holds for every $n \geq n_0$.

This works like a row of dominoes: the base case knocks over the first domino, and the inductive step shows that whenever one domino falls, the next one falls as well. Therefore, every domino in the row falls.

Definition 7. *(Rational numbers)*

So far, we only really know about rational numbers $\mathbb{Q}$. We want to build towards defining $\mathbb{R}$.

3 Sequence and limits

3.1 Decimal expansion

All of us in elementry or middle school have learnt about decimal expansion. For instance, we can think about $1/3 = 0.33333\ldots = 0.\bar{3}$. In general, we have learnt that every rational number $\frac{a}{b}$ has a decimal expansion. We learnt how to get this decimal expansion by finding remainder of a on division by b. We learn to do this process more carefully below through a recursive process:
Example 1: Note that $0.333\ldots$ can be though of using an infinite sum $0 + \frac{3}{10} + \frac{3}{10^2} + \frac{3}{10^3} + \ldots$.

So, the right way of thinking about decimal expansion of a number $\frac{a}{b}$ is trying to understand how to write $\frac{a}{b}$ as $\sum_{i \geq 0} \frac{a_i}{10^i}$ where $0 \leq a_i \leq 9$. Think about why do we need this bound on a_i?

We now explain the algorithm:
Let

$$x = \frac{a}{b},$$

where $a, b \in \mathbb{Z}$ and $b > 0$. By the Division Algorithm, there exist unique integers q and r_0 such that

$$a = bq + r_0, \qquad 0 \leq r_0 < b.$$

Thus

$$x = q + \frac{r_0}{b}.$$

It therefore suffices to construct a decimal expansion for $\frac{r_0}{b}$, which lies in $[0, 1)$.

Since $0 \leq r_0 < b$, we have

$$0 \leq 10r_0 < 10b.$$

Applying the Division Algorithm to $10r_0$, there exist unique integers d_1 and r_1 such that

$$10r_0 = bd_1 + r_1, \qquad 0 \leq r_1 < b.$$

Since $0 \leq 10r_0 < 10b$, the quotient d_1 satisfies

$$0 \leq d_1 \leq 9.$$

Dividing by $10b$, we obtain

$$\frac{r_0}{b} = \frac{d_1}{10} + \frac{r_1}{10b}.$$

Repeating the same argument for r_1, there exist unique integers d_2 and r_2 such that

$$10r_1 = bd_2 + r_2, \qquad 0 \leq r_2 < b,$$

and again

$$0 \leq d_2 \leq 9.$$

Substituting,

$$\frac{r_0}{b} = \frac{d_1}{10} + \frac{d_2}{10^2} + \frac{r_2}{b10^2}.$$

Continuing inductively, suppose r_n has been defined and satisfies

$$0 \leq r_n < b.$$

Since $0 \leq 10r_n < 10b$, the Division Algorithm yields unique integers d_{n+1} and r_{n+1} such that

$$10r_n = bd_{n+1} + r_{n+1}, \qquad 0 \leq r_{n+1} < b,$$

with

$$0 \leq d_{n+1} \leq 9.$$

Repeated substitution gives

$$\frac{r_0}{b} = \frac{d_1}{10} + \frac{d_2}{10^2} + \cdots + \frac{d_n}{10^n} + \frac{r_n}{b10^n}.$$

Since $0 \leq r_n < b$,

$$0 \leq \frac{r_n}{b10^n} < \frac{1}{10^n}.$$

Therefore

$$\left| \frac{r_0}{b} - \left(\frac{d_1}{10} + \frac{d_2}{10^2} + \cdots + \frac{d_n}{10^n} \right) \right| < \frac{1}{10^n}.$$

As $n \to \infty$, the right-hand side tends to 0. Hence

$$\frac{r_0}{b} = \sum_{n=1}^{\infty} \frac{d_n}{10^n}.$$

Thus

$$x = q + \sum_{n=1}^{\infty} \frac{d_n}{10^n},$$

which we denote by

$$q.d_1 d_2 d_3 \cdots .$$

This is the decimal expansion of x. **Example:** Consider the rational number

$$\frac{2}{11}.$$

Since

$$0 \leq 2 < 11,$$

we may take

$$r_0 = 2.$$

Applying the Division Algorithm to $10r_0 = 20$, we obtain

$$20 = 11 \cdot 1 + 9.$$

Thus

$$d_1 = 1, \qquad r_1 = 9,$$

and therefore

$$\frac{2}{11} = \frac{1}{10} + \frac{9}{11 \cdot 10}.$$

Next, we apply the same procedure to r_1. Since

$$10r_1 = 90,$$

the Division Algorithm gives

$$90 = 11 \cdot 8 + 2.$$

Thus

$$d_2 = 8, \qquad r_2 = 2,$$

and

$$\frac{2}{11} = \frac{1}{10} + \frac{8}{10^2} + \frac{2}{11 \cdot 10^2}.$$

At this point something important happens:

$$r_2 = 2 = r_0.$$

In other words, after two steps we have returned to exactly the same remainder with which we started.

Recall that at every stage we determine the next digit and next remainder by applying the Division Algorithm to $10r_n$:

$$10r_n = 11d_{n+1} + r_{n+1}, \qquad 0 \leq r_{n+1} < 11.$$

Since the quotient and remainder in the Division Algorithm are unique, the next digit and next remainder are completely determined by the current remainder.

Because $r_2 = r_0 = 2$, the calculation beginning with r_2 must be identical to the calculation beginning with r_0. Indeed,

$$10r_2 = 20 = 11 \cdot 1 + 9,$$

so

$$d_3 = 1, \qquad r_3 = 9.$$

But now

$$r_3 = 9 = r_1,$$

so the next step must be identical to the one that produced d_2 and r_2:

$$10r_3 = 90 = 11 \cdot 8 + 2,$$

giving

$$d_4 = 8, \qquad r_4 = 2.$$

Again,

$$r_4 = 2 = r_2,$$

so the same two calculations repeat forever. The sequence of remainders is therefore

$$2, 9, 2, 9, 2, 9, \ldots$$

and the corresponding sequence of digits is

$$1, 8, 1, 8, 1, 8, \ldots.$$

Hence

$$\frac{2}{11} = \frac{1}{10} + \frac{8}{10^2} + \frac{1}{10^3} + \frac{8}{10^4} + \cdots$$

and so

$$\frac{2}{11} = 0.181818\ldots = 0.\overline{18}.$$

Notice that the repetition of the digits is not a coincidence. It occurs because a remainder has repeated. Once a remainder repeats, all future digits and remainders must repeat as well, since each step of the construction is uniquely determined by the current remainder. Given a rational number $\frac{a}{b}$, either its decimal expansion stops after finitely many places or it repeats itself after a certain number of digits.

Problem 20. *Write out the decimal expansion of* $1/7, 1/13$ *using the above algorithm. Show that repetition after a certain number of decimals occurs in these cases as well. Can you explain why such a repetition must occur in the decimal expansion of* $\frac{a}{b}$ *for any integer* a, b*? (Hint: there are only finitely many numbers between 0 and 9.)*

From the above problem, we know that a number like $0.12345678\ldots$ can not be rational as the decimal expansion does not have any repetition. But $0.1 = \frac{1}{10}$ is a rational number, so is $0.12 = \frac{12}{100}$, so is $0.123 = \frac{123}{1000}$ and so on. This means if you mark the straight line with these numbers in order, there is a kind of hole you are missing. Which is the number $0.12345678\ldots$. To get the real numbers from rational numbers, you need to fill up so many gaps and you don't know how many there are. This needs a notion of "analytic completion". Therefore, getting Real numbers from rational numbers has not been a very easy process historically.

3.2 Discussing limits rigorously

Definition 8. *We say a sequence of rational numbers* $\{a_n\}$ *converges to* a *if for every rational number* $\epsilon > 0$*, there exists some* $N \in \mathbb{N}$ *so that if* $m > N$ *then* $|a - a_m| < \epsilon$*.*

Important points to discuss:

1. What is the point of using such a definition? Why can't you choose a fixed ϵ for instance $1, 0.1$ or 0.00001. Is there a point of allowing any $\epsilon > 0$?

2. To understand a complicated looking definition in mathematics, we should try to ask, what would it mean for the definition to not be true. What is the correct way to negate a statement? All of us intuitively understand that a sequence $\{a_n\}$ does not converge to a if you can find an $\epsilon > 0$ such that there does not exist N satisfying the property that for every $m > N$, $|a - a_m| < \epsilon$.

3. **Problem 21.** *(Back to Grandi's series) Consider the series defined by: $s_1 = 1$, $s_2 = 1 - 1$, $s_3 = 1 - 1 + 1$, and so on. Show that $\{s_n\}$ does not converge to 1 or 0.*

If a sequence has a limit, can it have two limits? Think pictorially first.

Theorem 2 (Archimedean Property of the Rationals)**.** *For any $m, n \in \mathbb{Q}$, if $m < n$ then $\exists c \in \mathbb{Q}$ such that $m < c < n$.*

Proof. We can just use Zeno's trick: given some choice of $m < n$, then we can try to find a point exactly halfway between them. Recall that the distance between m and n is given by the function $|m - n|$. It should follow that the midpoint between the two is given by $m + \frac{|m-n|}{2}$. Let's try to show this actually satisfies the property we care about:

1. **Claim 1:** $m < m + \frac{|m-n|}{2}$: Note that $0 < |m - n|$ by construction, and so $0 < \frac{|m-n|}{2}$ by multiplying $\frac{1}{2}$ to both sides. The claim follows by adding m to both sides.

2. **Claim 2:** $m + \frac{|m-n|}{2} < n$: First, note that $n = m + |m - n|$ (You should check this!). Thus, this claim is equivalent to showing that $m + \frac{|m-n|}{2} < m + |m - n|$. Subtracting m from both sides, we are left to show that $\frac{|m-n|}{2} < |m - n|$, which you can show from the definition by multiplying both sides by 2 and directly using the definition of the order relation on the natural numbers.

Thus, we have that $c = \frac{|m-n|}{2}$ satisfies the property. □

In a strong sense, the Archimedean property says that there is no such thing as an "infinitely small number" in the rationals: between 0 and any positive rational m, you can always find a smaller rational number between the two. This realization gives us our first important tool for this section:

Corollary 1. *For every $m, n \in \mathbb{Q}$, if for every $\epsilon > 0$ one has that $|m - n| < \epsilon$, then $m = n$.*

To show this, we'll have to introduce a, known as a *proof by contradiction.*

Proof of 1. Assume, towards a contradiction, that there is some pair $m, n \in \mathbb{Q}$ satisfying $\forall \epsilon > 0, |m - n| < \epsilon$ and that $m \neq n$. We will demonstrate that this

assumption leads to a contradiction.

Claim: There is some c for which $0 < c < |m - n|$.

Proof of claim: Because $m \neq n$, it must be the case that $m > n$ or $m < n$, using the fact that $<$ is an order relation on $\mathbb{Q}$. Thus, either $m - n > 0$ or $n - m > 0$, and thus $|m - n| > 0$. However, by the Archimedean Property of the rationals, there must be some $c \in \mathbb{Q}$ satisfying $0 < c < |m - n|$, yielding the claim.

However, we had already assumed that $\forall \epsilon > 0, |m - n| < \epsilon$, and the claim above proves the negation of this sentence. By contradiction, it could not be the case that $\forall \epsilon > 0, |m - n| < \epsilon$ and $m \neq n$. This concludes the proof. □

Theorem 3 (Uniqueness of Limits)**.** *Given a convergent sequence of rational numbers $\{a_n\}$ with limits $a, b \in \mathbb{Q}$, it must be the case that $a = b$. That is, if a sequence of rational numbers converges, it converges to only one limit.*

Proof. Let $\epsilon > 0$ be any choice. The following two statements we have by assumption:

- $\exists N_1 \in \mathbb{N}$ so that $\forall m \in \mathbb{N}$, if $m > N_1$ then $|a - a_m| < \frac{\epsilon}{2}$.
- $\exists N_2 \in \forall \epsilon > \not\vdash, \mathbb{N}$ so that $\forall m \in \mathbb{N}$, if $m > N$ then $|b - a_m| < \frac{\epsilon}{2}$.

Set $N = \max(N_1, N_2)$, where the notation "$\max(-,-)$"" means the largest of the inputs. Then the above two statements collectively imply:

- $\forall m \in \mathbb{N}$, if $m > N$ then $|a - a_m| < \frac{\epsilon}{2}$ **and** $|b - a_m| < \frac{\epsilon}{2}$.

Now, let's use the triangle inequality:

$$|a - b| \leq |a - a_m| + |a_m - b| = |a - a_m| + |b - a_m| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

In particular, reading off the start and the end, we've shown that $|a - b| < \epsilon$. However, $\epsilon > 0$ was just some random choice, and the above argument should work for *any* choice of $\epsilon > 0$. Really, what we've shown is:

- $\forall \epsilon > 0$, $|a - b| < \epsilon$.

And hence $a = b$ by our argument above. □

3.2.1 Back to our scheduled program on the Dichotomy Paradox

Recall that our resolution to trying to compute the "supertask" involved in Zeno's Dichotomy Paradox was to instead show that the following infinite collection of *finite* subtasks gets arbitrarily close to L as we compute larger and

larger portions:

$$
\begin{aligned}
s_0 &= \frac{L}{2} \\
s_1 &= \frac{L}{2} + \frac{L}{4} \\
s_2 &= \frac{L}{2} + \frac{L}{4} + \frac{L}{8} \\
&\dots \\
s_n &= \sum_{k=0}^{n} \frac{L}{2^{k+1}}
\end{aligned}
$$

Using the definition of limits, we would like to show that $\lim_{n\to\infty} s_n = L$. How do we do that?
First of all, in this specific case, we intuitively know that the limit must be L. Hence, we must show that for every rational number $\epsilon > 0$, we can find some N such that whenever $n > N$, $|L - s_n| < \epsilon$. First let's see if we can do it for some particular kind of arbitrary rational ϵ and proceed from there.

Claim: For every $n \in \mathbb{N}$, $L - s_n = \frac{L}{2^{n+1}}$.

Proof. Note that we're making an argument about infinitely many tasks indexed over the naturals, and we have exactly one tool to use here: the principle of induction.

- **Base case:** $L - s_0 = L - \frac{L}{2} = \frac{L}{2}$ and so the base case holds.
- **Inductive case:** Suppose we know already that $L - s_n = \frac{L}{2^{n+1}}$. We need to show that given this fact, $L - s_{n+1} = \frac{L}{2^{n+1}}$. Let's make the following simplifications.

$$
\begin{aligned}
L - s_{n+1} &= L - (s_n + \frac{L}{2^{n+2}}) \quad \text{(Expanding out the summation notation)} \\
L - (s_n + \frac{L}{2^{n+2}}) &= L - s_n - \frac{L}{2^{n+2}} \quad \text{(Fact about taking negatives)} \\
L - s_n - \frac{L}{2^{n+2}} &= \frac{L}{2^{n+1}} - \frac{L}{2^{n+2}} \quad \text{(By the inductive hypothesis)} \\
\frac{L}{2^{n+1}} - \frac{L}{2^{n+2}} &= \frac{2L}{2^{n+2}} - \frac{L}{2^{n+2}} = \frac{1}{2^{n+2}}(2L - L) \\
&= \frac{L}{2^{n+2}}
\end{aligned}
$$

  which concludes the inductive case, showing that the case for n implies the case for $n + 1$.

This concludes the proof of the intermediate claim above.

Let's now try to show the convergence directly. Namely, we need to show that given any $\epsilon > 0$, we need to show that there is an $N \in \mathbb{N}$ so that $m > N$ implies that $|L - s_m| < \epsilon$. Henceforth, let $\epsilon > 0$ be any choice. Using the claim above, we know that $|L - s_m| = \frac{L}{2^{m+1}}$. It follows that we just need to show that there is an N so that $\frac{L}{2^{m+1}} < \epsilon$ for every $m > N$.

Note that since ϵ is a rational number, we have that $\epsilon = \frac{a}{b}$ for some integers $a, b \in \mathbb{Z}$. Moreover, since $\epsilon > 0$, we have that $a \cdot b > 0$. This breaks us into two possible cases.

- **Case 1:** $a, b > 0$. In this case, we need to find an N so that $\frac{L}{2^{m+1}} < \frac{a}{b}$ for any $m > N$. We'll use the following steps.
  - For any $n \in \mathbb{N}$, using the scaling rule for the order relation, we know that $\frac{L}{2^{n+1}} < \frac{a}{b}$ if and only if $b \cdot \frac{L}{2^{n+1}} < b \cdot \frac{a}{b} = a$.
  - Similarly, we know that $b \cdot \frac{L}{2^{n+1}} < a$ if and only if $b \cdot L < 2^{n+1} \cdot a$.
  - Given a, b bigger than 0, there is always some $N \in \mathbb{N}$ so that $b \cdot L < 2^{N+1} \cdot a$. Furthermore, for any $m > N$, it must be the case that $b \cdot L << 2^{N+1} \cdot a < 2^{m+1} \cdot a$.

  It follows that in this case, there exists some N for which any $m > N$ satisfies $|L - s_m| = \frac{L}{s_m} < \frac{a}{b} = \epsilon$.
- **Case 2:** $a, b < 0$. Exercise!

As we have demonstrated the above for any particular choice of $\epsilon > 0$, we may conclude. □

Problem 22. *Argue using induction and ϵ, δ proof that*

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$$

for all $0 \leq r < 1$.

3.3 Algebraic limit theorem

Theorem 4. *Given convergent sequences a_n and b_n with limits a and b, the following are true:*

1. *Given any c, $\lim_{n \mapsto \infty}(c \cdot a_n) = c \cdot a$*
2. *$\lim_{n \mapsto \infty}(a_n + b_n) = a + b$*
3. *$\lim_{n \mapsto \infty}(a_n \cdot b_n) = a \cdot b$*
4. *$\lim_{n \mapsto \infty}\left(\frac{a_n}{b_n}\right) = \frac{a}{b}$.*

Proof. We will see a similar proof later. □

3.4 Cauchy sequence

But sometimes, a sequence might converge to some number, just not in the correct set. For example, a sequence of rational numbers might converge to an irrational number.

Theorem 5. *There does not exist a rational number a such that $a^2 = 2$.*

Proof. Assume, towards a contradiction, that such an a exists. Then $a \in \mathbb{Q}$, so there are integers $m, n \in \mathbb{Z}$ so that $a = \frac{m}{n}$, and we may furthermore assume that m and n have no common factors by reducing to lowest terms.

By assumption, $(\frac{m}{n})^2 = \frac{m^2}{n^2} = 2$, so $m^2 = 2n^2$. Since m and n have no common factors, we have that 2 divides m^2 and does not divide m. However, since 2 is prime, it must divide m itself, and so $2^2 = 4$ must divide m^2. However: $m^2 = 2n^2$, so we have shown that 4 must divide $2n^2$ and hence 2 must divide n. This is a contradiction: m and n were selected to have no common factors!

Thus, $a \in \mathbb{Q}$ satisfying $a^2 = 2$ could not have existed. In other words, there is no rational square root of 2. □

But definitely, such a number like $\sqrt{2}$ must make sense in real life. You can definitely plot it on the real line. Think about a right angle triangle with the two sides that are not the hypotenuse having length 1!

Proposition 1. *There is a sequence of positive rational numbers $\{a_n\}$ which does not converge to a rational number, but which satisfies $\lim_{n\to\infty}\{a_n^2\} = 2$.*

Proof. This will be developed in one of the problems given below. □

Discussion: Why did I write the above as a sequence of rational numbers whose square goes to 2 rather than a sequence of rational number that tends to $\sqrt{2}$?

Definition 9. *We say that a sequence $\{a_n\}$ is a Cauchy sequence if it satisfies the following: for all rational $\epsilon > 0$, there exists an $N \in \mathbb{N}$ so that for every $m, n > N$, we have that $|a_m - a_n| < \epsilon$.*

Intuitive understanding: Basically, a Cauchy sequence is one whose terms eventually get arbitrarily close together. Unwrapping the definition, for any maximum distance between the terms epsilon, if you go far enough into the sequence (past a number N), all pairs of terms (a_m, a_n) will be within that maximum distance of each other.

Example 1. *The condition*

$$|a_{n+1} - a_n| \longrightarrow 0$$

is not sufficient for a sequence to be Cauchy.

Consider the sequence

$$a_n = \sqrt{n}.$$

Then

$$\begin{aligned} |a_{n+1} - a_n| &= \sqrt{n+1} - \sqrt{n} \\ &= \frac{1}{\sqrt{n+1} + \sqrt{n}} \longrightarrow 0. \end{aligned}$$

Thus, consecutive terms of the sequence become arbitrarily close to one another.

However, the sequence is not Cauchy. Indeed, given any $N \in \mathbb{N}$, take $n = N$ and $m = 4N$. Then

$$|a_m - a_n| = \sqrt{4N} - \sqrt{N} = \sqrt{N}.$$

In particular, for $N > 1$ we have

$$|a_m - a_n| > 1.$$

Thus, no matter how far out in the sequence we go, we can find two terms whose distance is greater than 1. Hence (a_n) is not Cauchy.

This shows that controlling the distance between consecutive terms is not enough: the Cauchy condition requires

$$|a_m - a_n| < \varepsilon \qquad \text{for all } m, n \geq N.$$

Definition 10. *We say a sequence of rational numbers $\{a_n\}$* diverges *if it is not Cauchy.*

Problem 23. ***Discussion:*** *What is the difference between a convergent sequence and a Cauchy sequence? Show that every convergent sequence is Cauchy. If every Cauchy sequence is not convergent, what can we do to make the two notions equivalent?*

To solve the problem imposed in the above discussion, it seems like we have to "artificially" add limits of every Cauchy sequence in rationals to make a bigger set that is called the real numbers. But, after all, this process that we call artificial was very much inspired by our intuition and examples.

Definition 11. *The real numbers are a set, denoted $\mathbb{R}$, satisfying the following properties.*

1. *The rational numbers are contained in $\mathbb{R}$, i.e. $\mathbb{Q} \subseteq \mathbb{R}$.*
2. *Addition and multiplication extend to binary operations on $\mathbb{R}$, while the order relation given by inequalities also extends to $\mathbb{R}$. In particular, the order relation on $\mathbb{Q}$ extends to $\mathbb{R}$, and so does the absolute value function.*
3. *Every real number is the limit of a sequence of rational numbers.*
4. *(Axiom of Completeness) Every Cauchy sequence converges in $\mathbb{R}$.*

Remark 1. *One can also define the set of Real numbers as the set of all Cauchy sequences $\{x_n\}$ with the condition that two Cauchy sequences $\{x_n\}_n = \{x'_n\}_n$ are said to be equal if for all $\epsilon > 0$, we can find some N such that whenever $m > N$, $|x_m - x_{m'}| < \epsilon$.*

Problem 24. *In the above definition, I very casually said that the binary operations extend. But how? One of the theorems we have discussed previously helps us answer this question rigorously. Which theorem and how? Explain.*

Definition 12. *We say a sequence $\{a_n\}$ is monotone increasing if $\forall n \in \mathbb{N}$, $a_n \leq a_{n+1}$, and monotone decreasing if $\forall n \in \mathbb{N}$, $a_{n+1} \leq a_n$. We say a sequence is simply monotone if it is either monotone increasing or decreasing.*

Definition 13. *We say that a sequence $\{a_n\}$ is bounded if there is a positive real number $M \in \mathbb{R}$ such that $0 \leq |a_n| \leq M$ for every $M \in \mathbb{R}$.*

Theorem 6 (Monotone Convergence Theorem)**.** *If a sequence of real numbers $\{a_n\}$ is bounded and monotone, then it must converge.*

Proof. By the Axiom of Completeness and 23, it suffices to show that the bounded monotone sequence $\{a_n\}$ is a Cauchy sequence. (**Discuss:** Why am I imposing Axiom of completeness here? Think, as we discuss the proof about why this statement would have been false if we just were restricted to $\mathbb{Q}$).

We'll first handle the case when $\{a_n\}$ is monotone increasing. In the other case, we can just multiply by -1 to get something which is increasing, find a limit of the resultant sequence, and then use the Algebraic Limit Theorem to multiply by -1 again.

Now assume towards a contradiction that $\{a_n\}$ isn't a Cauchy sequence. Then there exists some $\epsilon > 0$ such that for any choice of $N \in \mathbb{N}$, there are $m > n > N$ satisfying $|a_m - a_n| > \epsilon$. Since we assumed the sequence was monotone increasing, we learn that $a_n + \epsilon < a_m$.

Thus, for a_0 there is an a_{n_1} with $a_0 + \epsilon < a_{n_1}$, and similarly for every a_{n_i} there is an $a_{n_{i+1}}$ satisfying $a_{n_i} + \epsilon < a_{n_{i+1}}$. In this way, we can construct an ascending sequence

$$a_0 < a_{n_1} < a_{n_2} < ... < a_{n_k} < ...$$

where $a_0 + k \cdot \epsilon < a_{n_k}$. Since $\epsilon > 0$, we may eventually choose k to be so large that $a_0 + k \cdot \epsilon$ is positive and also $a_0 + k \cdot \epsilon > M$; however, this is a contradiction, since we constructed a_{n_k} in the sequence satisfying $a_{n_k} > a_0 + k \cdot \epsilon$, but by the assumption of boundedness, every element of the sequence had absolute value less than M.

It follows that the sequence must be Cauchy, and hence that it converges. □

Theorem 7. *Every Cauchy sequence is bounded.*

Proof. Let (a_n) be a Cauchy sequence. Taking $\varepsilon = 1$ in the definition of a Cauchy sequence, there exists $N \in \mathbb{N}$ such that

$$|a_m - a_n| < 1$$

for all $m, n \geq N$. In particular,

$$|a_n - a_N| < 1$$

for all $n \geq N$. Hence, by the triangle inequality,

$$|a_n| = |a_n - a_N + a_N| \leq |a_n - a_N| + |a_N| < 1 + |a_N|$$

for all $n \geq N$.

Now set

$$M = \max\{|a_1|, \ldots, |a_{N-1}|, 1 + |a_N|\}.$$

Then $|a_n| \leq M$ for every $n \in \mathbb{N}$. Thus (a_n) is bounded. □

3.5 Bolzano-Weistrass

Let's examine some consequences of the monotone convergence theorem.

Proposition 2 (Direct Comparison Test)**.** *Let a_n, b_n be sequences of* positive *real numbers. Assume moreover that $\forall v \in \mathbb{N}, a_n \leq b_n$. Then we have the following.*

1. *If the series $\sum_{k=0}^{\infty} b_k$ converges (i.e., if the sequence of partial sums $s_n = \sum_{k=0}^{n} b_k$ converges), then the series $\sum_{k=0}^{\infty} a_k$ converges.*
2. *If the series $\sum_{k=0}^{\infty} a_k$ diverges (i.e., if the sequence of partial sums $t_n = \sum_{k=0}^{n} a_k$ diverges), then the series $\sum_{k=0}^{\infty} b_k$ diverges.*

Proof. For part 1, note that since every element of the sequence $\{a_k\}$ is positive, the sequence $t_n = \sum_{k=0}^{n} a_k$ is monotone increasing, and similarly the sequence $s_n = \sum_{k=0}^{n} b_k$ is monotone increasing. We realise the following two facts.

1. Set $s := \lim_{n\to\infty} s_n = \lim_{n\to\infty}(\sum_{k=0}^{n} b_k)$. It is not hard to see that $\forall k, s_k \leq s$; the sequence s_k is monotone increasing, and thus the limit is always greater than any individual finite sum (Check this formally!).
2. Since we assumed that $\forall k, a_k \leq b_k$, we have that $\forall n, t_n \leq s_n$ and thus t_n is less than s.

Since t_n is bounded above by s and monotone, we may apply Theorem 6 and conclude.

For Part 2, let's use contradiction. Suppose $\sum_{k=0}^{\infty} b_k$ converges. Then prom part 1, $\sum_{k=0}^{\infty} a_k$ must converge, which leads us to a contradiction. □

Example 2 (Harmonic Series)**.** *Let's use the above to show that the following series diverges:*

$$\{a_k\} = \frac{1}{k}\ ,\ \ s_n = \sum_{k=1}^{n} a_k = \sum_{k=0}^{n} \frac{1}{k}$$

which is known as the harmonic series. We'll do this by direct comparison: for any positive number n, the number $2^{\lceil \log_2 n \rceil}$ represents the smallest natural

number power of 2 *that is still larger than* n.[1] *In particular, we have the relation* $n \leq 2^{\lceil \log_2 n \rceil}$. *Let's define the sequence*

$$\{b_n\} := \frac{1}{2^{\lceil \log_2 n \rceil}}$$

which, when written out explicitly starting from $n = 1$, *is the sequence* $\{1, \frac{1}{2}, \frac{1}{4}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, ...\}$.

By the discussion above, it's clear that for every n, $b_n \leq a_n$ *because* $\frac{1}{2^{\lceil \log_2 n \rceil}} \leq \frac{1}{n}$. *In order to show that* $\sum_{k=0}^{n} \frac{1}{n}$ *diverges, it will be sufficient by the Direct Comparison Test to show that* $t_n := \sum_{k=0}^{n} \frac{1}{2^{\lceil \log_2 n \rceil}}$ *diverges.*

Why is this latter fact true? Well, notice the following pattern:

$$\begin{aligned} t_1 &= 1 + \frac{1}{2} \\ t_4 &= 1 + \frac{1}{2} + \frac{1}{2} = 1 + 1 = 2 \\ t_8 &= 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = 1 + 1 + 1 = 3 \\ &... \end{aligned} \tag{1}$$

We see a pattern and this is a ripe place to use induction. Here afterwards, it is an exercise!

Proposition 3. *Every infinite sequence has a monotone subsequence.*

Discussion. Before proving the proposition, let us think about what an arbitrary sequence might look like. Imagine plotting the points

$$(1, a_1), (2, a_2), (3, a_3), \ldots$$

from left to right. Starting at some point, suppose that we can always find another point farther to the right which is lower than the one we are currently at. Then we can simply keep moving to the right and downward, producing a decreasing subsequence.

So the interesting case is when this process gets stuck. What does it mean to get stuck? It means that we have reached some a_k for which every term to its right lies above it:

$$a_k \leq a_l \qquad \text{for every } l > k.$$

We will call such a term a *valley*.

Now imagine that there is no decreasing subsequence. Then, no matter how far to the right we go, we must eventually encounter a valley; otherwise we could keep moving to the right and downward forever. Consequently, there must be infinitely many valleys.

[1] The notation $\lceil x \rceil$ is the *ceiling* function, and is the function sending a real number x to the smallest integer that is greater than x; the identification in the sentence above should follow.

But the valleys have a very useful property. If a_{k_1} and a_{k_2} are two valleys with $k_1 < k_2$, then a_{k_2} lies to the right of a_{k_1}, and hence

$$a_{k_1} \leq a_{k_2}.$$

Thus, as we move from one valley to the next, their heights can only increase. The valleys themselves therefore form an increasing subsequence.

This suggests a dichotomy: either we can keep moving downward forever, giving a decreasing subsequence, or we cannot, in which case the resulting valleys give an increasing subsequence.

Proof. Let (a_n) be an infinite sequence. Suppose that (a_n) has no monotone decreasing subsequence.

We first claim that for every $N \in \mathbb{N}$, there exists some $k \geq N$ such that

$$a_k \leq a_l \qquad \text{for every } l > k.$$

Indeed, suppose otherwise. Then for some N, every $k \geq N$ admits an $l > k$ such that

$$a_l < a_k.$$

Starting with $n_1 = N$, we may therefore choose successively

$$n_1 < n_2 < n_3 < \cdots$$

such that

$$a_{n_1} > a_{n_2} > a_{n_3} > \cdots.$$

This is a monotone decreasing subsequence, contradicting our assumption. The claim follows.

We now construct a sequence of valleys. By the claim, choose k_1 such that

$$a_{k_1} \leq a_l \qquad \text{for every } l > k_1.$$

Applying the claim with $N = k_1 + 1$, choose $k_2 > k_1$ with the same property. Continuing inductively, we obtain

$$k_1 < k_2 < k_3 < \cdots$$

such that each a_{k_i} is a valley.

Since $k_{i+1} > k_i$ and a_{k_i} is a valley, we have

$$a_{k_i} \leq a_{k_{i+1}}$$

for every i. Hence

$$a_{k_1} \leq a_{k_2} \leq a_{k_3} \leq \cdots,$$

so (a_{k_i}) is a monotone increasing subsequence.

Thus, if (a_n) has no monotone decreasing subsequence, it has a monotone increasing subsequence. Therefore, every infinite sequence has a monotone subsequence. □

Remark 2. *Logically, the above proof proceeded as follows. There are two cases:*
1. Either there is a monotone decreasing sequence.
2. Or there isn't.
If case 1 is true, we are done.
If Case 2 is true, we produced a monotone increasing subsequence.

Theorem 8. *Any bounded sequence has a convergent subsequence.*

Proof. Any sequence has a monotone subsequence from previous theorem. Now since this given sequence is bounded, its subsequence will also be bounded. Now, from Monotone convergence theorem, we are done! □

Problem 25. *(Section)[Density of the Rationals] Using the axiomatic characterization of the real numbers, show that for any real numbers $a, b \in \mathbb{R}$ with $a < b$, there is a rational number r satisfying*

$$a < r < b.$$

Problem 26. *Suppose that a is a rational number and r is an irrational number. Prove that $a + r$ is irrational. (No need to try to do what we were doing with $\sqrt{2}$. Think in terms of representing r in terms of a Cauchy sequence. Can you use some sort of contradiction to prove this?)*

Problem 27 (Cesàro Means)**.** *Show that if $\{x_n\}$ is a convergent sequence, then the sequence given by the averages*

$$y_n = \frac{x_1 + x_2 + x_3 + \cdots + x_n}{n}$$

is convergent, and converges to the same limit.

Problem 28. *Let $a'_n = |a_n|$, and let its associated sequence of partial sums be denoted by*

$$s'_n = \sum_{k=0}^{n} |a_k|.$$

Show that if s'_n converges, then s_n also converges (although not necessarily to the same limit). [Note: You cannot use Monotone convergence theorem here. Why? But triangle inequality is available, however it only applies to finitely many terms. So be careful!!]

Definition 14. *Let $f : \mathbb{R} \to \mathbb{R}$ be a function. We say f is a* contraction mapping *if there exists a constant $0 < C < 1$ such that, for every pair $x, y \in \mathbb{R}$,*

$$|f(x) - f(y)| < C|x - y|.$$

Problem 29. *Given a contraction mapping f, show that for every $x \in \mathbb{R}$, the sequence*

$$\{x, f(x), f(f(x)), \ldots, f^{(n)}(x), \ldots\}$$

converges. Then show that the resulting limit

$$L := \lim_{n\to\infty} f^{(n)}(x)$$

satisfies

$$f(L) = L.$$

This is known as a fixed point *of f.*

Remark 3. *What's the point of the above? We want to find a fixed point of f by using the following general idea:*

$$f \circ f^{(\infty)}(x) = f^{(\infty+1)}(x) = f^{(\infty)}(x).$$

This is another example of an infinite supertask, this time not involving addition: we tried to "apply f infinitely many times." Of course, this isn't a well-defined thing that you or I could do. Instead, we reasoned about the infinite sequence of finite tasks, which involved composing f finitely many times.

To caution you that supertasks usually have bad answers unless you assume something nice, let's do the following.

Problem 30. *Give an example of a function $f : \mathbb{R} \to \mathbb{R}$ and an input value $x \in \mathbb{R}$ for which the sequence*

$$\{x, f(x), f(f(x)), \ldots\}$$

does not converge. If you haven't already, give an example for which this corresponding sequence is bounded, but still does not converge to a limit.

Problem 31 (Heron's Method)**.**

- *Let $x_0 \in [1, 2]$, and define $f : [1, 2] \to \mathbb{R}$ as $f(x) = \frac{1}{2}\left(x + \frac{2}{x}\right)$*

  $$x_{n+1} = \frac{1}{2}\left(x_n + \frac{2}{x_n}\right).$$

  Using the contraction mapping result above, show that the sequence $\{x_n\}$ converges and find its limit.

- *The above was a contraction mapping only when the source was $[1, 2]$. It is not a contraction mapping if you define the function $f : [1, \infty) \to \mathbb{R}$. What fails?*

4 Oral presentation: A tussle between logic and intuition in mathematics

Consider the definition of existence of a limit.

$$\text{For all } \epsilon > 0, \exists \delta \text{ such that blah blah}$$

What does the negation of such a statement mean? You might very easily be able to come up with an answer. Now consider the following statement:

It is not true that I did not think that choosing to do math will be totally not impossible

Observe:
"totally not impossible" = not impossible,
"I did not think ..." = another negation,
"It is not true that ..." = a third negation.
In classical logic, two negations cancel, so three negations reduce to one negation. Negation simply reverses the truth value of a proposition.
So

It is not true that I did not think that choosing to do math will be totally not impossible

simplifies to roughly

I did not think that choosing to do math would be impossible.

Equivalently, in more natural English:

I thought that choosing to do math was possible.

These examples illustrate that a careful study of formal mathematic is not just important but makes our life easy in math.

The Law of Excluded Middle: Every (mathematical) statement is either true or false, and not both.

4.1 Propositional Calculus

Mathematical arguments rely on a small collection of logical rules and conventions. For example, when we use phrases such as "P or Q" or "if P, then Q," we must agree on exactly what these phrases mean.

In this section, we will study basic logical operations using truth tables. Once these definitions are established, we can use them to verify standard logical identities, such as De Morgan's laws. This systematic use of truth tables is called **propositional calculus**.

4.1.1 Equivalence

Let P and Q be statements. We say that P and Q are **equivalent** if they always have the same truth value. Thus, whenever P is true, Q is also true, and whenever P is false, Q is also false.

We write $P \equiv Q$ to indicate that P and Q are equivalent.

The truth table for equivalence is

P	Q	$P \equiv Q$
T	T	T
T	F	F
F	T	F
F	F	T

The four rows represent all possible combinations of truth values of P and Q.

Here is an example of two equivalent mathematical statements. Let

$$P: \quad n \text{ is an even integer,}$$

and let

$$Q: \quad n^2 \text{ is an even integer.}$$

Then $P \equiv Q$.

Indeed, if n is even, then $n = 2k$ for some integer k, and hence $n^2 = 4k^2$, so n^2 is even.

Conversely, if n^2 is even, then n must be even. For if n were odd, we could write $n = 2k + 1$, and then

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1,$$

which is odd. Thus n^2 being even implies that n is even.

Therefore the two statements always have the same truth value, and hence

$$n \text{ is even} \quad \equiv \quad n^2 \text{ is even.}$$

4.1.2 Negation

For a statement P, we write $\neg P$ for the negation of P. It is read "not P," and we think of it as the assertion that P is false. The truth table defining negation is:

P	$\neg P$
T	F
F	T

Proposition 4. *Let P be a statement. Then*

$$\neg(\neg P) \equiv P.$$

Proof. We construct a truth table, in which both sides of $\equiv$ appear as columns, and then we compute the columns.

P	$\neg P$	$\neg(\neg P)$
T	F	T
F	T	F

Since the columns corresponding to the RHS (right hand side) and LHS (left hand side) agree, we conclude that the statements are equivalent. □

4.1.3 Conjunction and Disjunction

Let P and Q be two statements. The statement

$$P \wedge Q$$

is called the **conjunction** of P and Q, and is read "P and Q." It is true precisely when both P and Q are true. Thus, conjunction allows us to combine two statements while requiring both of them to hold.

The statement

$$P \vee Q$$

is called the **disjunction** of P and Q, and is read "P or Q." In mathematics, the word "or" is inclusive: $P \vee Q$ is true when P is true, when Q is true, or when both are true. This differs from the way "or" is sometimes used in everyday language, where it may mean that exactly one of the two alternatives occurs.

The definitions of conjunction and disjunction can be summarized by the following truth table:

P	Q	$P \wedge Q$	$P \vee Q$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	F

Thus, $P \wedge Q$ is true only when both statements are true, whereas $P \vee Q$ is false only when both statements are false.

4.1.4 Implication

We explain using truth table what $P \implies Q$ means:

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Problems that help us become precise in math:

1. Prove that $(P \implies Q) \equiv Q \vee (\neg P)$.
   Using the above, find the negation of the ϵ, δ definition. The idea is to understand the ϵ, δ definition using a propositional logic statement.

2. In the definition 11, I very casually mentioned that the order relation extends from $\mathbb{Q}$ to $\mathbb{R}$ without being precise. The point of this exercise is to do that. If we view elements $x, y \in \mathbb{R}$ as a Cauchy sequences $\{x_n\}$ and $\{y_n\}$, formulate a definition of $\{x_n\} < \{y_n\}$.
   Verify that your definition is correct using the following metric: Suppose, a, b are two rational numbers such that $\{a_n\}$ converges to a and $\{b_n\}$ converges to b. Then

   $$a < b \text{ as per the definition of inequality in rationals } \iff \{a_n\} < \{b_n\}$$

5 Functions

Definition 15. *(Function) A function is a collection of pairs of numbers with the following property: if (a, b) and (a, c) are both in the collection, then $b = c$; in other words, the collection must not contain two different pairs with the same first element. number.*

Definition 16. *(Domain) If f is a function, the **domain** of f is the set of all a for which there is some b such that (a, b) is in f. If a is in the domain of f, it follows from the definition of a function that there is, in fact, a unique number b such that (a, b) is in f. This unique b is denoted by $f(a)$.*

Hence, we can express a function as $f : A \to B$

Problem 32. *Is the following rule a function?*

$$f(x) = \begin{cases} x^2 + 1, & x \geq 0, \\ -x, & x \leq 0. \end{cases}$$

Why or why not?

Definition 17. *(Polynomial functions)*

Definition 18. *(Rational functions)*

Problem 33. *Let $f(x) = \dfrac{1}{x-1}$. Define $f^{(2)}(x) = f(f(x))$, which means applying f to x 2 times. What are the minimum number of points you need to remove from $\mathbb{R}$ for f to be a well defined function on $\mathbb{R}$? What about $f^{(2)}$? What would those numbers be in both the cases?*

5.1 Graphs

We will learn plotting graphs through examples

1. $f(x) = mx + b$
2. $f(x) = x^2, x^3$
3. $f(x) = x^2 - x$
4. $f(x) = \dfrac{1}{x^2+1}$
5. Assuming we know what $\sin x$ means and that its zeroes occur at the numbers $x = n\pi$, 2π being thought of the number that defines the angle going full way around a circle.
6. $\sin \dfrac{1}{x}$

7. $x \sin \dfrac{1}{x}$

8. $f(x) = |x|$, $f(x) = |x + 5|$. By definition $|y| = +y$ if $y \geq 0$ or $|y| = -y$ if $y \leq 0$.

Intersection of two graphs

Definition 19. *(Circle) A circle with center (a, b) and radius $r > 0$ contains, by definition, all the points (x, y) whose distance from (a, b) is equal to r. That is why an equation of circle is given by all points (x, y) that satisfy $(x - a)^2 + (y - b)^2 = r^2$. Note that this does not really define a function of y dependent on x as for every value of x there are two different values of y. You could however restrict it to be a function by defining*

$$y(x) = \sqrt{r^2 - (x - a)^2} + b$$

or

$$y(x) = -\sqrt{r^2 - (x - a)^2} + b$$

by taking the convention that the symbol $\sqrt{}$ means taking the positive square root.

Problem 34. *Recall the Pythagorean theorem: if a triangle has side lengths x, y, and z, and the sides of lengths x and y meet at an angle of $\frac{\pi}{2}$, then*

$$x^2 + y^2 = z^2.$$

Conversely, if a triangle with side lengths x, y, and z satisfies

$$x^2 + y^2 = z^2,$$

then the sides of lengths x and y meet at an angle of $\frac{\pi}{2}$.

Let

$$f(x) = mx + b, \qquad g(x) = nx + c.$$

Prove that the graphs of f and g are perpendicular whenever

$$mn = -1.$$

(Hint: First determine the point of intersection of the two graphs, then use the Pythagorean theorem by computing appropriate squared distances.)

Problem 35. *Recall the triangle inequality $|x + y| \leq |x| + |y|$. Using this, prove the following inequality:*

$$\sqrt{(x_1 + y_1)^2 + (x_2 + y_2)^2} \leq \sqrt{x_1^2 + x_2^2} + \sqrt{y_1^2 + y_2^2}.$$

for any numbers x_1, x_2, y_1, y_2 (Hint: Reinterpret these numbers as points on a plane)

Problem 36. *Describe the graph of* $g(x) := max(x^2 - x, 0)$

Problem 37. *For a real number* x*, let* $\lfloor x \rfloor$ *denote the greatest integer less than or equal to* x*. For example,*

$$\lfloor 2.1 \rfloor = 2, \qquad \lfloor -0.9 \rfloor = -1.$$

For each of the following functions, determine its domain and sketch its graph.

5.

$$f(x) = \left\lfloor \frac{1}{x} \right\rfloor.$$

6.

$$f(x) = \frac{1}{\left\lfloor \frac{1}{x} \right\rfloor}.$$

Problem 38. *There is a well known theorem that there exists a unique circle passing through three non colinear points (i.e. three points not in the same line). Prove this theorem in a special case. Find the equation of the circle passing through the three points*

$$(1, 0), \qquad (0, 1), \qquad (2, 2).$$

. Using the general form of equation of a circle, explain what goes wrong if three given points $P_1 = (x_1, y_1)$*,* $P_2 = (x_2, y_2)$*,* $P_3 = (x_3, y_3)$ *are colinear. Explain every step.*

6 Limits

> "And what are these same evanescent increments? They are neither finite quantities, nor quantities infinitely small, nor yet nothing. May we not call them **the ghosts of departed quantities**?"
>
> — George Berkeley

The concept of "limits" is the most important in all of calculus. Everything we study from now on will be based of this definition of limit. **An imprecise definition:** The function f approaches the limit l near a if we can make $f(x)$ as close to l as we want by getting closer and closer to a, but unequal to a.
Remark: The words "near" or "closer and closer but not equal to" are important above. Consider the graph of a function that looks as follows:

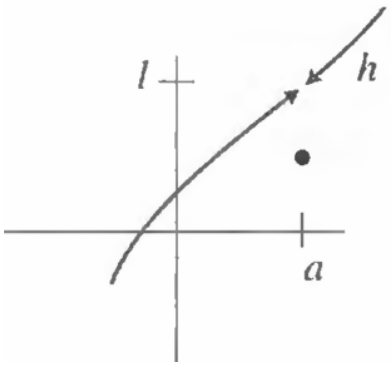


Figure 1: Figure 1

Even though the value of $f(a) \neq l$, the limit is l. How do we understand this mathematically? First of all note that getting closer and closer to a but not being equal to a means $|x-a|$ is not 0 but it is almost approaching 0. Moreover taking modulus is also important as x can approach a from both directions. The same thing holds true for $|f(a)-l|$.

Definition 20. *We write*

$$\lim_{x \to a} f(x) = \ell$$

and say that f approaches the limit ℓ as x approaches a *if the following condition holds:*

For every $\varepsilon > 0$*, there exists a* $\delta > 0$ *such that for all* x*,*

$$0 < |x - a| < \delta$$

implies

$$|f(x) - \ell| < \varepsilon.$$

How do we use the above definition to prove mathematical statements about f approaching certain l as x approaches a?
First thing to do that is to realize that for any given ϵ, you have to produce a δ, which means in some sense this δ is supposed to be a function of ϵ. Remember that ϵ can be any number bigger than 0, it can be 100, 1, 0.01, 0.00000785, even $\pi/3.14$. So the way to prove such statement using this $\epsilon - \delta$ definition is to assume first that $|f(x) - l| < \epsilon$ and then produce a δ satisfying the above property. We discuss a few examples below:

1. Proving that if $f(x) = 9x + 3$, then $f(x)$ approaches 21 as x approaches 2. Choose an $\epsilon > 0$. To show what we need, suppose that $|f(x) - 21| < \epsilon$, we need to show that there is some $\delta = \delta(\epsilon)$ such that $|x - 2| < \delta$. Now $|f(x) - 21| = |9x + 3 - 21| = |9x - 18| = |9(x - 2)| = 9|x - 2| < \epsilon$. Hence, taking $\delta = \epsilon/9$ works! And we are done.

Linear functions give a very nice expression of δ in terms of ϵ. How do we deal with more complicated functions?

2. Suppose $f(x) = x^3$. We want to show that $\lim_{x\to 2} x^3 = 8$, this means that for every $\epsilon > 0$, we need to find a δ (depending on ϵ), such that whenever $|x^3 - 8| < \epsilon$, $|x - 2| < \delta$. Note $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$. Before we start doing this for general ϵ, let us take $\epsilon = 0.2$. We want to find the correct δ for which $|x - 2| < \delta$ would mean that $|x^3 - 8| < 0.2$.If you look at the figure below, it says that x should be between 2.017 and 1.983. Now let $\delta_1 = |2.016 - 2| = 0.016$ and $\delta_2 = |1.983 - 2| = 0.017$. How, what is the correct δ to choose? Choose the smaller one as it is stronger. So when $|x - 2| < 0.016$, $1.984 < x < 2.016$. You can check from the figure that the interval $(1.984, 2.016)$ is within the correct bound and so when $|x - 2| < \delta = 0.016$, we have $|x^3 - 8| < 0.2$.

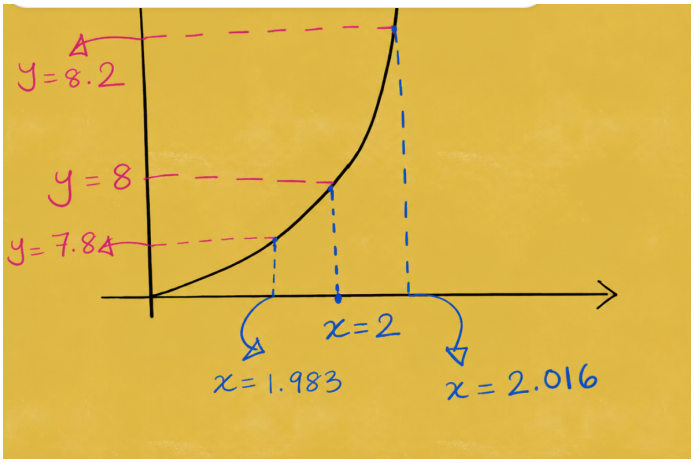


Figure 2: x^3 near $x = 2$

But then, I need to deal with general ϵ as well. Suppose I set $|x - 2| < 1$ for now. This means that the maximum value of $(x^2 + 2x + 4)$ is less than $1 + 2 + 4 = 7$. Now, $|(x - 2)||(x^2 + 2x + 4)| < 7|x - 2|$. What if I take $|x - 2| < \epsilon/7 \implies |x^3 - 8| = |(x - 2)||(x^2 + 2x + 4)| < \epsilon$. So $\delta = \epsilon/7$ should suffice. But we need to be careful! What if $\epsilon/7 > 1$. Therefore, I choose $\delta = \min(1, \epsilon/7)$.

3. Let us deal with the function $f(x) = \sqrt{2x + 1}$ now. We want to show that $\lim_{x\to 4} \sqrt{2x + 1} = 9$. You can deal with the specific case of $\epsilon = 0.2$ again using the picture below. The correct δ to choose will be $min(0.58, 0.62)$, which is 0.58.

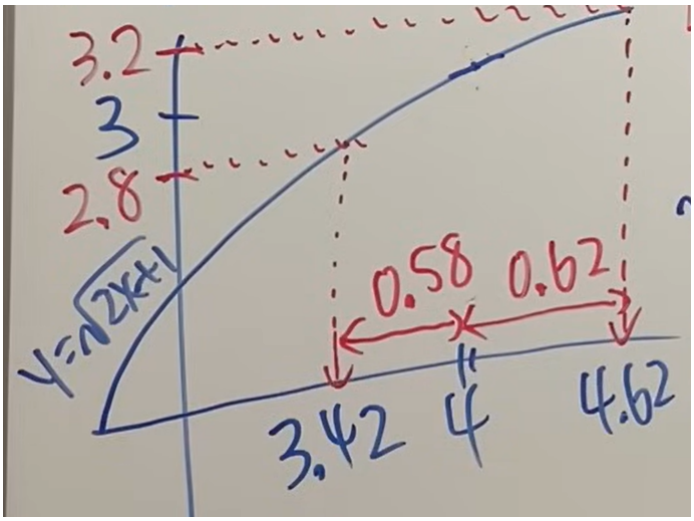


Figure 3: $\sqrt{2x+1}$ near $x = 4$

But now let us write the general proof.

Proof. Given $\epsilon > 0$, choose

$$\delta = \frac{\epsilon}{2}.$$

Suppose

$$0 < |x - 4| < \delta.$$

Then

$$\begin{aligned} \left|\sqrt{2x+1} - 3\right| &= \left|\frac{(\sqrt{2x+1} - 3)(\sqrt{2x+1} + 3)}{\sqrt{2x+1} + 3}\right| \\ &= \left|\frac{2x + 1 - 9}{\sqrt{2x+1} + 3}\right| \\ &= \frac{2|x-4|}{\sqrt{2x+1} + 3}. \end{aligned}$$

Since

$$\sqrt{2x+1} + 3 > 1,$$

we have

$$\left|\sqrt{2x+1} - 3\right| < 2|x - 4|.$$

Since $|x - 4| < \delta$, it follows that

$$\left|\sqrt{2x+1} - 3\right| < 2\delta = \epsilon.$$

Therefore,

$$\lim_{x \to 4} \sqrt{2x+1} = 3.$$

□

Discuss: What was the difference in our approach towards choosing the correct δ between example 1 and 2?

Notice an important difference between this example and the previous one. For $f(x) = x^3$, after factoring we had

$$|x^3 - 8| = |x - 2||x^2 + 2x + 4|.$$

Although we can make $|x-2|$ small by choosing δ, the second factor $|x^2+2x+4|$ also depends on x. We therefore first restricted x to a small fixed interval around 2, by requiring $|x - 2| < 1$. This gave us a constant upper bound for the second factor. For $f(x) = \sqrt{2x+1}$, something nicer happens. After rationalizing,

$$\left|\sqrt{2x+1} - 3\right| = \frac{2|x-4|}{\sqrt{2x+1}+3}.$$

This time the extra expression involving x appears in the denominator. We do not need to know exactly how large the denominator is. We only need a lower bound for it, and we automatically have

$$\sqrt{2x+1} + 3 \geq 3.$$

Hence

$$\left|\sqrt{2x+1} - 3\right| \leq \frac{2}{3}|x-4|.$$

Thus, unlike in the previous example, we do not first need to impose something like $|x - 4| < 1$. We already have the estimate we need.

The goal in an ϵ-δ proof is not necessarily to "solve for δ" directly. The goal is to find a constant C such that, whenever x is sufficiently close to a,

$$|f(x) - L| \leq C|x - a|.$$

Then, if we choose

$$|x - a| < \frac{\epsilon}{C},$$

we get

$$|f(x) - L| < \epsilon.$$

Sometimes, as in the example $f(x) = x^3$, we first need to restrict x to a small interval around a in order to find such a constant C. Sometimes, as in the example $f(x) = \sqrt{2x+1}$, we can find such a constant immediately.

Problem 39. *Prove that* $\lim_{x\to 2} \dfrac{1}{x} = \dfrac{1}{2}$.

Discussion:
We proved a certain limit exists for a linear polynomial, for the cubic function x^3 and from your exercise, even for $\dfrac{1}{x}$. Each case involved a different computation of δ. This hints at the existence of a uniform way that helps us compute limit of functions by breaking them down into smaller sub pieces.

Theorem 9. *A function cannot approach two different limits near a. In other words, if f approaches ℓ near a, and f approaches m near a, then $\ell = m$.*

Proof. Since this is our first theorem about limits it will certainly be necessary to translate the hypotheses according to the definition.

Since f approaches ℓ near a, we know that for any $\epsilon > 0$ there is some number $\delta_1 > 0$ such that, for all x,

$$0 < |x - a| < \delta_1 \quad \Longrightarrow \quad |f(x) - \ell| < \epsilon.$$

We also know, since f approaches m near a, that there is some $\delta_2 > 0$ such that, for all x,

$$0 < |x - a| < \delta_2 \quad \Longrightarrow \quad |f(x) - m| < \epsilon.$$

We have had to use two numbers, δ_1 and δ_2, since there is no guarantee that the δ which works in one definition will work in the other. But, in fact, it is now easy to conclude that for any $\epsilon > 0$ there is some $\delta > 0$ such that, for all x,

$$0 < |x - a| < \delta \quad \Longrightarrow \quad |f(x) - \ell| < \epsilon \quad \text{and} \quad |f(x) - m| < \epsilon;$$

we simply choose

$$\delta = \min(\delta_1, \delta_2).$$

To complete the proof we just have to pick a particular $\epsilon > 0$ for which the two conditions

$$|f(x) - \ell| < \epsilon \qquad \text{and} \qquad |f(x) - m| < \epsilon$$

cannot both hold, if $\ell \neq m$. The proper choice is suggested by the figure. If $\ell \neq m$, so that $|\ell - m| > 0$, we can choose

$$\epsilon = \frac{|\ell - m|}{2}.$$

It follows that there is a $\delta > 0$ such that, for all x,

$$0 < |x - a| < \delta \quad \Longrightarrow \quad |f(x) - \ell| < \frac{|\ell - m|}{2}$$

and

$$|f(x) - m| < \frac{|\ell - m|}{2}.$$

This implies that for $0 < |x - a| < \delta$ we have

$$\begin{aligned} |\ell - m| &= |\ell - f(x) + f(x) - m| \\ &\leq |\ell - f(x)| + |f(x) - m| \\ &< \frac{|\ell - m|}{2} + \frac{|\ell - m|}{2} \\ &= |\ell - m|, \end{aligned}$$

which is a contradiction. Therefore $\ell = m$. □

Theorem 10. *If*

$$\lim_{x\to a} f(x) = \ell \qquad \textit{and} \qquad \lim_{x\to a} g(x) = m,$$

then

1. $\lim_{x\to a}(f+g)(x) = \ell + m$;
2. $\lim_{x\to a}(f\cdot g)(x) = \ell \cdot m$.

Moreover, if $m \neq 0$, then

3. $\lim_{x\to a}\left(\frac{1}{g}\right)(x) = \frac{1}{m}$.

Proof. The hypothesis means that for every $\epsilon > 0$ there are $\delta_1, \delta_2 > 0$ such that, for all x,

$$0 < |x-a| < \delta_1 \quad \Longrightarrow \quad |f(x) - \ell| < \epsilon,$$

and

$$0 < |x-a| < \delta_2 \quad \Longrightarrow \quad |g(x) - m| < \epsilon.$$

This means (since, after all, $\epsilon/2$ is also a positive number) that there are $\delta_1, \delta_2 > 0$ such that, for all x,

$$0 < |x-a| < \delta_1 \quad \Longrightarrow \quad |f(x) - \ell| < \frac{\epsilon}{2},$$

and

$$0 < |x-a| < \delta_2 \quad \Longrightarrow \quad |g(x) - m| < \frac{\epsilon}{2}.$$

Now let $\delta = \min(\delta_1, \delta_2)$. If $0 < |x-a| < \delta$, then $0 < |x-a| < \delta_1$ and $0 < |x-a| < \delta_2$ are both true, so both

$$|f(x) - \ell| < \frac{\epsilon}{2} \qquad \text{and} \qquad |g(x) - m| < \frac{\epsilon}{2}$$

are true. Hence

$$\begin{aligned} |(f+g)(x) - (\ell + m)| &= |(f(x) - \ell) + (g(x) - m)| \\ &\leq |f(x) - \ell| + |g(x) - m| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} \\ &= \epsilon. \end{aligned}$$

This proves (1).

To prove (2), we proceed similarly. If $\epsilon > 0$, there are $\delta_1, \delta_2 > 0$ such that, for all x,

$$0 < |x-a| < \delta_1 \quad \Longrightarrow \quad |f(x) - \ell| < \min\left(1, \frac{\epsilon}{2(|m|+1)}\right),$$

and

$$0 < |x - a| < \delta_2 \quad \Longrightarrow \quad |g(x) - m| < \frac{\epsilon}{2(|\ell| + 1)}.$$

Again let $\delta = \min(\delta_1, \delta_2)$. If $0 < |x - a| < \delta$, then

$$|f(x) - \ell| < \min\left(1, \frac{\epsilon}{2(|m| + 1)}\right) \qquad \text{and} \qquad |g(x) - m| < \frac{\epsilon}{2(|\ell| + 1)}.$$

In particular, $|f(x) - \ell| < 1$, so

$$|f(x)| \leq |f(x) - \ell| + |\ell| < |\ell| + 1.$$

Therefore,

$$\begin{aligned} |f(x)g(x) - \ell m| &= |f(x)(g(x) - m) + m(f(x) - \ell)| \\ &\leq |f(x)|\,|g(x) - m| + |m|\,|f(x) - \ell| \\ &< (|\ell| + 1)\frac{\epsilon}{2(|\ell| + 1)} + |m|\frac{\epsilon}{2(|m| + 1)} \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} \\ &= \epsilon. \end{aligned}$$

This proves (2).

Finally, suppose $m \neq 0$. If $\epsilon > 0$, there is a $\delta > 0$ such that, for all x,

$$0 < |x - a| < \delta \quad \Longrightarrow \quad |g(x) - m| < \min\left(\frac{|m|}{2}, \frac{\epsilon |m|^2}{2}\right).$$

The first inequality gives

$$|g(x)| \geq |m| - |g(x) - m| > \frac{|m|}{2} > 0,$$

so $g(x) \neq 0$. Moreover,

$$\begin{aligned} \left|\frac{1}{g(x)} - \frac{1}{m}\right| &= \frac{|m - g(x)|}{|m|\,|g(x)|} \\ &< \frac{2|g(x) - m|}{|m|^2} \\ &< \epsilon. \end{aligned}$$

Thus

$$\lim_{x \to a} \left(\frac{1}{g}\right)(x) = \frac{1}{m}.$$

This proves (3). □

Problem 40. *Suppose the functions f and g have the following property: for all $\epsilon > 0$ and all x,*

$$\text{if } 0 < |x-2| < \sin^2\left(\frac{\epsilon^2}{9}\right) + \epsilon, \text{ then } |f(x)-2| < \epsilon,$$

and

$$\text{if } 0 < |x-2| < \epsilon^2, \text{ then } |g(x)-4| < \epsilon.$$

For each $\epsilon > 0$ find a $\delta > 0$ such that, for all x,

(i) if $0 < |x-2| < \delta$, then

$$|f(x) + g(x) - 6| < \epsilon.$$

(ii) if $0 < |x-2| < \delta$, then

$$|f(x)g(x) - 8| < \epsilon.$$

(iii) if $0 < |x-2| < \delta$, then

$$\left|\frac{1}{g(x)} - \frac{1}{4}\right| < \epsilon.$$

(iv) if $0 < |x-2| < \delta$, then

$$\left|\frac{f(x)}{g(x)} - \frac{1}{2}\right| < \epsilon.$$

Problem 41. *Compute the following limits from first principles (i.e. ϵ, δ definition:*

1. $\lim_{x\to 1} \dfrac{1-\sqrt{x}}{1-x}$
2. $\lim_{x\to 0} x(3 - \cos(x^2))$
3. $\lim_{x\to 0} x \sin\dfrac{1}{x}$

Problem 42. *Give example to show that the following definition of $\lim_{x\to a} f(x) = l$ is not correct:*

(a) For all $\delta > 0$ there is an $\epsilon > 0$ such that if $0 < |x-a| < \delta$, then

$$|f(x) - l| < \epsilon.$$

7 Continuous functions

Let $A, B \subseteq \mathbb{R}$ and let

$$f : A \longrightarrow B.$$

Let $a \in A$. We say that f is *continuous at* a if for every $\epsilon > 0$, there exists $\delta > 0$ such that

$$f\big((a-\delta, a+\delta) \cap A\big) \subseteq (f(a)-\epsilon, f(a)+\epsilon) \cap B.$$

In other words, whenever

$$x \in (a-\delta, a+\delta) \cap A,$$

we have

$$f(x) \in (f(a)-\epsilon, f(a)+\epsilon) \cap B.$$

Equivalently, for every $\epsilon > 0$, there exists $\delta > 0$ such that for every $x \in A$,

$$|x-a| < \delta \quad \Longrightarrow \quad |f(x)-f(a)| < \epsilon.$$

Theorem 11. *Let $f : A \to \mathbb{R}$ be a function, and let $c \in A$. Then f is continuous at c if and only if for every sequence $\{a_n\} \subseteq A$ which converges to c,*

$$\lim_{n\to\infty} f(a_n) = f(c).$$

Proof. First suppose that f is continuous at c, and let $\{a_n\}$ be a sequence in A converging to c. We want to show that

$$\lim_{n\to\infty} f(a_n) = f(c).$$

Fix $\epsilon > 0$. Since f is continuous at c, there exists $\delta > 0$ such that

$$|x-c| < \delta \quad \Longrightarrow \quad |f(x)-f(c)| < \epsilon.$$

Since $a_n \to c$, there exists $N \in \mathbb{N}$ such that whenever $n > N$,

$$|a_n - c| < \delta.$$

Therefore, whenever $n > N$,

$$|f(a_n)-f(c)| < \epsilon.$$

Hence

$$\lim_{n\to\infty} f(a_n) = f(c).$$

We now prove the other direction by contrapositive. Suppose that f is not continuous at c. Then there exists some $\epsilon > 0$ such that for every $\delta > 0$, there exists $x \in A$ satisfying

$$|x-c| < \delta \qquad \text{and} \qquad |f(x)-f(c)| > \epsilon.$$

We will construct a sequence $\{x_n\} \subseteq A$ which converges to c, but for which $\{f(x_n)\}$ does not converge to $f(c)$.

For each $n \in \mathbb{N}$, take

$$\delta = \frac{1}{2^{n+1}}.$$

By our assumption, there exists $x_n \in A$ such that

$$|x_n - c| < \frac{1}{2^{n+1}}$$

and

$$|f(x_n) - f(c)| > \epsilon.$$

Since

$$\frac{1}{2^{n+1}} \to 0,$$

we have

$$x_n \to c.$$

On the other hand,

$$|f(x_n) - f(c)| > \epsilon$$

for every n. Thus the sequence $\{f(x_n)\}$ stays at least a fixed distance ϵ away from $f(c)$, and therefore

$$f(x_n) \not\to f(c).$$

Thus, if f sends every sequence converging to c to a sequence converging to $f(c)$, then f must be continuous at c. □

7.1 The Least Upper Bound Property of the Reals

Definition 21 (Least Upper Bounds)**.** *Let $A \subseteq \mathbb{R}$ be a nonempty subset of real numbers.*

1. *We say that A admits an upper bound if there exists some $M \in \mathbb{R}$ such that for every $x \in A$, $x \leq M$. Any choice of such an M is called an upper bound of A.*
2. *We say that $M \in \mathbb{R}$ is a least upper bound for A if it is an upper bound of A, and for every upper bound N of A, $M \leq N$.*

The following trick is a very useful reformulation of the property of being a least upper bound.

Proposition 5. *Let $A \subseteq \mathbb{R}$ be some nonempty subset of the real numbers. $M \in \mathbb{R}$ is a least upper bound for A if and only if:*

1. *M is an upper bound of A.*
2. *For every $\epsilon > 0$, there exists an $a \in A$ such that*

$$M - \epsilon < a.$$

Proof. Let us show the forward direction first: namely, if M is a least upper bound of A, then it satisfies conditions (1) and (2) above.

$$M \text{ is a least upper bound of } A \implies$$

- M satisfies condition (1): This is part of the definition of being a least upper bound.
- M satisfies condition (2): Fix $\epsilon > 0$. Since M is a least upper bound, $M - \epsilon$ cannot be an upper bound of A, since $M - \epsilon < M$. Thus, there must exist some $a \in A$ such that

$$M - \epsilon < a.$$

It remains to show the backward direction: namely, that satisfying conditions (1) and (2) of the proposition is enough to imply being a least upper bound. To this end, let $M \in \mathbb{R}$ be some element satisfying conditions (1) and (2).

$$M \text{ satisfies conditions (1) and (2)} \implies M \text{ is a least upper bound of } A.$$

By assumption, M is an upper bound of A. It remains to show that if $N \in \mathbb{R}$ is any other upper bound of A, then $M \leq N$.

By contraposition, this amounts to showing that if $N < M$, then N cannot be an upper bound of A. If $N < M$, then $N = M - \epsilon$ for some $\epsilon > 0$, and condition (2) implies that there exists $x \in A$ such that

$$N = M - \epsilon < x.$$

Thus, N cannot be an upper bound unless it is greater than or equal to M. □

Corollary 2. *Let $A \subseteq \mathbb{R}$ be some nonempty subset of the real numbers, and let M be an upper bound of A. Then M is a least upper bound if and only if there exists some sequence $\{a_n\}$ of elements of A such that*

$$\lim_{n \to \infty} a_n = M.$$

Proof. Let us first show the forward direction: namely, let us assume that M is a least upper bound, and try to build this sequence by hand.

$$M \text{ is a least upper bound} \implies \text{there is a sequence in } A \text{ converging to } M.$$

From Proposition 3.2.3, we know that M is a least upper bound if and only if for every $\epsilon > 0$, there exists an $a \in A$ such that

$$M - \epsilon < a.$$

Using this, we can build a sequence by hand as follows:

$$\begin{aligned}
&\text{Let } a_1 \in A \text{ be selected so that } M - 1 < a_1 \leq M.\\
&\text{Let } a_2 \in A \text{ be selected so that } M - \frac{1}{2} < a_2 \leq M.\\
&\text{Let } a_3 \ldots\\
&\text{Let } a_n \in A \text{ be selected so that } M - \frac{1}{n} < a_n \leq M.
\end{aligned}$$

We claim that the constructed sequence a_n converges to M. Fix any $\epsilon > 0$. There exists some $N \in \mathbb{N}$ such that

$$0 < \frac{1}{N} < \epsilon.$$

By construction, for every $n > N$, we have

$$M - \frac{1}{n} < a_n \leq M,$$

and hence

$$|M - a_n| < \epsilon.$$

Such a selection worked for any arbitrary choice of $\epsilon > 0$, and thus this sequence must converge to M.

We now show the other direction, namely that if M is an upper bound and the limit of a sequence in A, then it is a least upper bound of A.

$$\begin{aligned}
&M \text{ is an upper bound of } A \text{ and } M \text{ is the limit of a sequence in } A\\
&\quad \Longrightarrow M \text{ is the least upper bound of } A.
\end{aligned}$$

We again invoke Proposition 3.2.3 here. Suppose that there exists a sequence $\{a_n\}$ of elements of A with limit M. Fix any $\epsilon > 0$. Convergence of a_n implies the weaker statement that there exists some a_n satisfying

$$|M - a_n| < \epsilon.$$

However, since M is an upper bound of A, it must be the case that

$$a_n \leq M.$$

Altogether, we have

$$M - \epsilon < a_n \leq M.$$

Since $\epsilon > 0$ was an arbitrary choice, we learn that for any $\epsilon > 0$ we may find an $a \in A$ such that

$$M - \epsilon < a \leq M.$$

By above proposition, we may conclude that M is a least upper bound. □

7.2 Intermediate value theorem using least upper bounds

Lemma 1. ?? *Let $a, b \in \mathbb{R}$ with $a \leq b$. Suppose that $\{a_n\}$ is a convergent sequence such that $a \leq a_n \leq b$ for every $n \in \mathbb{N}$. If $\lim_{n\to\infty} a_n = L$, then $L \in [a, b]$. Equivalently, $a \leq L \leq b$.*

Proof. We prove that the limit cannot lie outside the interval $[a, b]$.

First suppose, for contradiction, that $L < a$. Let $\epsilon = \frac{a-L}{2} > 0$. Since $a_n \to L$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $|a_n - L| < \epsilon$. Thus $a_n < L + \epsilon = L + \frac{a-L}{2} = \frac{a+L}{2} < a$. This contradicts the assumption that $a_n \geq a$ for every n.

Similarly, suppose for contradiction that $L > b$. Let $\epsilon = \frac{L-b}{2} > 0$. Since $a_n \to L$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $|a_n - L| < \epsilon$. Thus $a_n > L - \epsilon = L - \frac{L-b}{2} = \frac{L+b}{2} > b$. This contradicts the assumption that $a_n \leq b$ for every n.

Therefore L is neither less than a nor greater than b. Hence $a \leq L \leq b$, so $L \in [a, b]$. □

Discussion: Why does the above fail for an open interval (a, b)?

Theorem 12 (Intermediate Value Theorem)**.** *Let $f : [a, b] \to \mathbb{R}$ be a continuous function. If $f(a) \neq f(b)$, then for any value d between $f(a)$ and $f(b)$, there is an element $c \in [a, b]$ such that*

$$f(c) = d.$$

Proof. If $d = f(a)$ or $d = f(b)$, then the result is immediate. Thus, we may assume that d lies strictly between $f(a)$ and $f(b)$. Assume, without loss of generality, that

$$f(a) < d < f(b).$$

Suppose, towards a contradiction, that there does not exist any $c \in [a, b]$ such that $f(c) = d$. Consider the set

$$A = \{x \in [a, b] : f(x) < d\}.$$

Since $f(a) < d$, we have $a \in A$, so A is nonempty. Also, $A \subseteq [a, b]$, and therefore b is an upper bound for A. Consequently, A has a least upper bound. Let

$$c = \sup A.$$

Because $a \leq c \leq b$, the point c belongs to the domain of f. We will show that $f(c) = d$.

Since c is the least upper bound of A, there exists a sequence $\{a_n\}$ of elements of A such that

$$\lim_{n\to\infty} a_n = c.$$

Indeed, for every $n \in \mathbb{N}$, the number $c - \frac{1}{n}$ cannot be an upper bound for A. Hence we may choose $a_n \in A$ such that

$$c - \frac{1}{n} < a_n \leq c.$$

It follows that $a_n \to c$.

Since f is continuous at c, we have

$$\lim_{n\to\infty} f(a_n) = f\left(\lim_{n\to\infty} a_n\right) = f(c).$$

On the other hand, $a_n \in A$ for every n, so

$$f(a_n) < d.$$

Therefore,

$$f(c) \leq d.$$

We now rule out the possibility that $f(c) < d$. Suppose that

$$f(c) < d.$$

Set

$$\epsilon = d - f(c) > 0.$$

By continuity of f at c, there exists $\delta > 0$ such that whenever $x \in [a, b]$ satisfies $|x - c| < \delta$, we have

$$|f(x) - f(c)| < \epsilon.$$

This implies that

$$f(x) - f(c) \leq |f(x) - f(c)| < \epsilon,$$

and hence

$$f(x) < f(c) + \epsilon = d.$$

Since $f(b) > d$ while $f(c) < d$, we have $c < b$. Choose

$$0 < h < \min\{\delta, b - c\}.$$

Then $c + h \in [a, b]$ and

$$|(c + h) - c| = h < \delta.$$

It follows that

$$f(c + h) < d.$$

Thus $c + h \in A$. But $c + h > c$, contradicting the fact that c is an upper bound for A.

Therefore, $f(c) < d$ is impossible. Since we already know that $f(c) \leq d$, we must have

$$f(c) = d.$$

This contradicts our assumption that no such c exists. Therefore, there is an element $c \in [a, b]$ such that

$$f(c) = d.$$

If $f(a) > d > f(b)$, we apply the preceding argument to the continuous function $-f$. Hence the result holds whenever d lies between $f(a)$ and $f(b)$. $\square$

Problem 43. *Use intermediate value theorem to show that any polynomial in odd degree has atleast 1 root in $\mathbb{R}$. Use the fact that for a large enough positive x, x^n will always be bigger than $a_1x^{n-1} + a_2x^{n-2} + \ldots + a_0$ for any given choice of real numbers a_i.*

Theorem 13. *(Boundedness theorem) Let $f : [a, b] \to \mathbb{R}$ be a continuous function. Then f is bounded, that is, there exists some $M \in \mathbb{R}$ such that $|f(x)| \leq M$. (Note: we don't necessarily need a least upper bound, finding some M such that $|f(x)| < M$ is also enough.)*

Before we do this proof, let us discuss some examples:

1. Does the function $f : (0, 1] \to \mathbb{R}$ defined by $x \mapsto \dfrac{1}{x}$ have a maximum?

2. If the conclusion from above example is that the interval $(0, 1]$ is not closed, what if I define a function Define $f : [0, 1] \to \mathbb{R}$ by

$$f(x) = \begin{cases} \dfrac{1}{x}, & 0 < x \leq 1, \\ 1, & x = 0. \end{cases}$$

   Now we have a closed interval, does this have a maximum? If yes, why, if not, what hypothesis of the Theorem 13 is failing?

Proof of Theorem 13:

Proof. Assume, towards a contradiction, that f is not bounded. Then for every $n \in \mathbb{N}$, there exists an element $x_n \in [a, b]$ satisfying $|f(x_n)| > n$. In this way, we may build a sequence $\{x_n\}$ of elements in $[a, b]$.

We now have two sequences $\{x_n\}$ and $\{f(x_n)\}$ If the sequence $\{x_n\}_n$ itself converged, we would be done: if x_n converged, then it had to converge to some $x \in [a, b]$, and by continuity we would have $\lim_{n\to\infty} f(x_n) = f(x)$. However, the sequence $f(x_n)$ is not bounded and thus has no limit, i.e. it cannot converge. Contradiction! It must be the case that no such convergent sequence existed.

However, we do not know whether or not $\{x_n\}$ converges. This is where Lemma **??** comes in: we know that $\{x_n\}$ has to have a convergent subsequence $\{x_{n_k}\}$. Moreover, by our assumption on the original sequence, we know that $|f(x_{n_k})| > n_k$. In particular, since n_k is an increasing sequence of natural numbers indexed over k, the sequence $f(x_{n_k})$ must not have been bounded.

Contradiction! If $\{x_{n_k}\}$ converges to some $x \in [a, b]$, then $f(x) = \lim_{k\to\infty} f(x_{n_k})$. However, a limit of the sequence $\{f(x_{n_k})\}$ cannot exist, as by construction it is not a bounded sequence.

□

In the previous theorem, we showed the existence of an upper bound M, but the question is whether some upperbound can be realized as one of the values of the function itself. And as long as we are in a closed interval, the answer is yes!

Theorem 14 (Extreme Value Theorem)**.** *Let $f : [a, b] \to \mathbb{R}$ be a continuous function. Then there exists a point $c \in [a, b]$ such that for every $x \in [a, b]$, $f(x) \leq f(c)$. Such a point will be called a maximum of the function f on $[a, b]$.*

Proof. Proof. We use the same sequence-construction argument. By the Boundedness Theorem, the range of f, given by

$$\text{Range}(f) = \{y \in \mathbb{R} : \text{ there exists } x \in [a, b] \text{ such that } f(x) = y\},$$

has an upper bound and therefore has a least upper bound. Denote this least upper bound by $\sup f$.

Our goal is to show that there exists $c \in [a, b]$ such that

$$f(c) = \sup f.$$

There exists a sequence $\{y_n\}$ in Range(f) such that

$$y_n \longrightarrow \sup f.$$

Since each y_n belongs to Range(f), there exists $x_n \in [a, b]$ such that

$$f(x_n) = y_n$$

for every n.

Because $\{x_n\}$ is a sequence in the closed and bounded interval $[a, b]$, it has a convergent subsequence $\{x_{n_k}\}$ whose limit L belongs to $[a, b]$. Since f is continuous,

$$f(L) = \lim_{k\to\infty} f(x_{n_k}).$$

Moreover, $\{f(x_{n_k})\}$ is a subsequence of the convergent sequence $\{f(x_n)\}$. Therefore,

$$\lim_{k\to\infty} f(x_{n_k}) = \lim_{n\to\infty} f(x_n) = \sup f.$$

Consequently,

$$f(L) = \sup f.$$

Thus, setting $c = L$, we obtain a point $c \in [a, b]$ such that $f(x) \leq f(c)$ for every $x \in [a, b]$. □

□

Problem 44. *For each of the intervals of the form $I_1 = [a, b)$, $I_2 = (a, b]$ and $I_3 = (a, b)$, give an example of 3 different continuous functions $f_k : I_k \to \mathbb{R}$ for k in the set $\{1, 2, 3\}$ such that f_k fails to have a maximum.*

7.3 Uniform continuity

Before we start discussing what the notion of a uniformly continuous functions mean, we recall the difference between a convergent sequence and a Cauchy sequence. Convergent sequences are the ones for which you check the ϵ, N definition with respect to a certain point, however, Cauchy sequences are the ones

where we use the ϵ, N definition to check distances between two different elements in the sequence. (Go back in the notes to check the precise difference).

We learnt before that a continuous function takes a convergent sequence to a convergent sequence, but does a continuous sequence take a Cauchy sequence to a Cauchy sequence?

Before we answer that question, let us discuss the definition of a uniformly continuous function.

Definition 22. *A function $f : A \to \mathbb{R}$ is* uniformly continuous *on A if, for every $\epsilon > 0$, there exists a $\delta > 0$ such that, for any pair $x, y \in A$ satisfying*

$$|x - y| < \delta,$$

we have

$$|f(x) - f(y)| < \epsilon.$$

Note the contrast with ordinary continuity:

- With ordinary continuity, we say that at a particular point $c \in A$, there is some choice of $\delta > 0$ such that

  $$|x - c| < \delta$$

  implies

  $$|f(x) - f(c)| < \epsilon.$$

  In some sense, this is a *local choice*: the choice of δ at some c may not work for other points in A.

- For uniform continuity, we demand that there be a choice of δ that works for every possible pair $x, y \in A$. This is a *global choice*.

Problem Set

Problem 45. *Let $A, B \subseteq \mathbb{R}$. Suppose that $f : A \to \mathbb{R}$ and $g : B \to \mathbb{R}$ are continuous. Prove that*

$$g \circ f : f^{-1}(B) \to \mathbb{R}$$

is continuous, where

$$f^{-1}(B) = \{x \in A : f(x) \in B\}.$$

Problem 46. *Show that every polynomial of odd degree has at least one real root.*

Problem 47. *Let $I \subseteq \mathbb{R}$ be an interval. Prove that every continuous one-to-one function*

$$f : I \to \mathbb{R}$$

is strictly monotone. That is, prove that f is either strictly increasing or strictly decreasing on I.

Hint: First show that if $x_1 < x_2 < x_3$ are in I, then $f(x_2)$ lies strictly between $f(x_1)$ and $f(x_3)$.

Problem 48. ***(*)*** *Let $I \subseteq \mathbb{R}$ be an interval, and suppose that $f : I \to \mathbb{R}$ is continuous and one-to-one. Define*

$$g : f(I) \to I$$

by

$$g(f(x)) = x.$$

Prove that g is continuous.

Problem 49. ***(*)*** *Let m and n be positive integers with $\gcd(m, n) = 1$. Define*

$$D_n = \begin{cases} \mathbb{R}, & \text{if } n \text{ is odd,} \\ [0, \infty), & \text{if } n \text{ is even.} \end{cases}$$

Prove that

$$f : D_n \to \mathbb{R}, \qquad f(x) = x^{m/n},$$

is continuous.

Hint: Consider the function $h_n(x) = x^n$ on D_n and use the preceding problem together with the composition problem.

8 Derivatives and integrals

Examples of $|x|$ and x^2.

Definition 23. ***The function f is differentiable at*** *a if*

$$\lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

exists.

In this case the limit is denoted by $f'(a)$ and is called the ***derivative*** *of f at a. We also say that f is* ***differentiable*** *if f is differentiable at a for every a in the domain of f.*

Another way of saying that a function is differentiable is asking that the limit $\lim_{y \to c} \dfrac{f(y) - f(c)}{y - c}$ exists.

Discussion: Can you think of a definition of continuity using converging sequences to c?

Problem 50. *Show that $f(x) = x^{1/3}$ is not differentiable at $x = 0$. Show that it is differentiable at all other points and find its derivative.*

Derivative as instantaneous velocity and tangent at a point on graph.

Definition 24. *Let f be defined on an interval containing a.*
*The **right-hand derivative** of f at a is*

$$f'_+(a) = \lim_{h\to 0^+} \frac{f(a+h)-f(a)}{h},$$

provided this limit exists.
*The **left-hand derivative** of f at a is*

$$f'_-(a) = \lim_{h\to 0^-} \frac{f(a+h)-f(a)}{h},$$

provided this limit exists.

Theorem 15. *If f is differentiable at a, then f is continuous at a.*

Proof.

$$\begin{aligned}\lim_{h\to 0}\big(f(a+h)-f(a)\big) &= \lim_{h\to 0}\left(\frac{f(a+h)-f(a)}{h}\cdot h\right)\\ &= \left(\lim_{h\to 0}\frac{f(a+h)-f(a)}{h}\right)\left(\lim_{h\to 0} h\right)\\ &= f'(a)\cdot 0\\ &= 0.\end{aligned}$$

□

Suppose $f : A \to \mathbb{R}$ is a function such that it is differentiable at every point of A. Then $f' : A \to \mathbb{R}$ is denoted as the derivative function. We will also denote $f'(c) = \dfrac{df}{dx}\,|_{x=c}$. (You can say out loud the latter object as evaluating $\dfrac{df}{dx}$ at the point $x = c$.)

8.1 Differentials of some common functions

Example 3. *Consider the function*

$$f(x) = x^2.$$

We prove that f is differentiable at every $a \in \mathbb{R}$. By definition,

$$\frac{f(a+h)-f(a)}{h} = \frac{(a+h)^2-a^2}{h} = \frac{2ah+h^2}{h} = 2a+h$$

for every $h \neq 0$. Therefore,

$$\lim_{h\to 0}\frac{f(a+h)-f(a)}{h} = \lim_{h\to 0}(2a+h) = 2a.$$

Hence f is differentiable at a, and

$$f'(a) = 2a.$$

More explicitly, given $\epsilon > 0$, choose $\delta = \epsilon$. If $0 < |h| < \delta$, then

$$\left| \frac{f(a+h) - f(a)}{h} - 2a \right| = |2a + h - 2a| = |h| < \delta = \epsilon.$$

Thus the difference quotient converges to $2a$ as $h \to 0$.

Example 4. *Now consider the function*

$$g(x) = |x|.$$

We show that g is not differentiable at 0. For $h \neq 0$, its difference quotient at 0 is

$$\frac{g(0+h) - g(0)}{h} = \frac{|h|}{h}.$$

If $h > 0$, then

$$\frac{|h|}{h} = 1,$$

whereas if $h < 0$, then

$$\frac{|h|}{h} = -1.$$

Thus, as h approaches 0 from the right, the difference quotient equals 1, while as h approaches 0 from the left, it equals -1.

We can see directly from the ϵ–δ definition that the limit cannot exist. Suppose, for contradiction, that there were some $L \in \mathbb{R}$ such that

$$\lim_{h \to 0} \frac{|h|}{h} = L.$$

Take $\epsilon = \frac{1}{2}$. Then there would exist $\delta > 0$ such that whenever

$$0 < |h| < \delta,$$

we would have

$$\left| \frac{|h|}{h} - L \right| < \frac{1}{2}.$$

Taking $h = \frac{\delta}{2} > 0$ gives

$$|1 - L| < \frac{1}{2},$$

while taking $h = -\frac{\delta}{2} < 0$ gives

$$|-1 - L| < \frac{1}{2}.$$

However, the triangle inequality would then imply

$$2 = |1 - (-1)| \leq |1 - L| + |L - (-1)| < \frac{1}{2} + \frac{1}{2} = 1,$$

which is impossible. Therefore, the difference quotient has no limit as $h \to 0$*, so* $g(x) = |x|$ *is not differentiable at* 0*.*

Example 5. *Consider the function*

$$f(x) = \begin{cases} x \sin\left(\dfrac{1}{x}\right), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

First, notice that f *is continuous at* 0*. Indeed,*

$$\left| x \sin\left(\frac{1}{x}\right) \right| \leq |x|,$$

so

$$\lim_{x \to 0} x \sin\left(\frac{1}{x}\right) = 0 = f(0).$$

However, f *is not differentiable at* 0*. Its difference quotient at* 0 *is*

$$\frac{f(0+h) - f(0)}{h} = \frac{h \sin(1/h)}{h} = \sin\left(\frac{1}{h}\right).$$

As $h \to 0$*, the number* $1/h$ *becomes arbitrarily large, and* $\sin(1/h)$ *continues to oscillate between* -1 *and* 1*.*

More precisely, suppose, for contradiction, that there exists $L \in \mathbb{R}$ *such that*

$$\lim_{h \to 0} \sin\left(\frac{1}{h}\right) = L.$$

Take $\epsilon = \frac{1}{2}$*. Then there would exist* $\delta > 0$ *such that*

$$0 < |h| < \delta$$

implies

$$\left| \sin\left(\frac{1}{h}\right) - L \right| < \frac{1}{2}.$$

Choose a sufficiently large positive integer n *so that both*

$$h_n = \frac{1}{\frac{\pi}{2} + 2\pi n} \qquad \text{and} \qquad k_n = \frac{1}{\frac{3\pi}{2} + 2\pi n}$$

are smaller than δ*. Then*

$$\sin\left(\frac{1}{h_n}\right) = \sin\left(\frac{\pi}{2} + 2\pi n\right) = 1$$

and

$$\sin\left(\frac{1}{k_n}\right) = \sin\left(\frac{3\pi}{2} + 2\pi n\right) = -1.$$

Therefore, the assumed limit would give

$$|1 - L| < \frac{1}{2}$$

and

$$|-1 - L| < \frac{1}{2}.$$

By the triangle inequality,

$$2 = |1 - (-1)| \leq |1 - L| + |L - (-1)| < 1,$$

which is impossible. Hence the difference quotient does not have a limit as $h \to 0$*, and therefore* f *is not differentiable at* 0*.*

Thus, this is an example of a function that is continuous at 0 *but not differentiable there.*

Definition 25 (Higher derivatives)**.** *Let* $f : I \to \mathbb{R}$ *be a function defined on an open interval* I*.*

The first derivative of f *is denoted by*

$$f^{(1)} = f'.$$

If f' *is differentiable, then its derivative is called the* second derivative *of* f *and is denoted by*

$$f^{(2)} = f'' = (f')'.$$

More generally, define

$$f^{(0)} = f.$$

For an integer $n \geq 1$*, if* $f^{(n-1)}$ *is differentiable, then the* nth derivative *of* f *is defined recursively by*

$$f^{(n)} = \left(f^{(n-1)}\right)'.$$

Thus, at a point $a \in I$*,*

$$f^{(n)}(a) = \lim_{h \to 0} \frac{f^{(n-1)}(a+h) - f^{(n-1)}(a)}{h},$$

provided this limit exists.

We say that f *is* n times differentiable *on* I *if* $f^{(1)}, f^{(2)}, \ldots, f^{(n)}$ *all exist on* I*.*

Problem 51. *Consider the function*

$$f(x) = \begin{cases} x^2 \sin\left(\dfrac{1}{x}\right), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Is it differentiable at $x = 0$*. (Check it carefully!)*

Problem 52. *Let $f : \mathbb{R} \to \mathbb{R}$ be a function satisfying*

$$|f(x)| \leq x^2$$

for every $x \in \mathbb{R}$. Prove that f is differentiable at 0.

Problem 53 (Linearity of the Derivative)**.** *Suppose that $f, g : (a, b) \to \mathbb{R}$ are both differentiable at $c \in (a, b)$. Prove the following statements.*

1. *The function $f + g$, defined by*

$$(f + g)(x) = f(x) + g(x),$$

   is differentiable at c, and

$$(f + g)'(c) = f'(c) + g'(c).$$

2. *For every constant $k \in \mathbb{R}$, the function kf, defined by*

$$(kf)(x) = kf(x),$$

   is differentiable at c, and

$$(kf)'(c) = kf'(c).$$

Theorem 16 (Product Rule)**.** *Suppose that $f, g : (a, b) \to \mathbb{R}$ are both differentiable at $c \in (a, b)$. Then the product fg, defined by*

$$(fg)(x) = f(x)g(x),$$

is differentiable at c, and

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c).$$

Proof. By the definition of the derivative,

$$(fg)'(c) = \lim_{x \to c} \frac{f(x)g(x) - f(c)g(c)}{x - c}.$$

We add and subtract $f(x)g(c)$ in the numerator to obtain

$$f(x)g(x) - f(c)g(c) = f(x)\big(g(x) - g(c)\big) + g(c)\big(f(x) - f(c)\big).$$

Therefore,

$$\begin{aligned}(fg)'(c) &= \lim_{x \to c} \left(f(x) \frac{g(x) - g(c)}{x - c} + g(c) \frac{f(x) - f(c)}{x - c} \right) \\ &= \left(\lim_{x \to c} f(x) \right) \left(\lim_{x \to c} \frac{g(x) - g(c)}{x - c} \right) + g(c) \left(\lim_{x \to c} \frac{f(x) - f(c)}{x - c} \right).\end{aligned}$$

Since differentiability implies continuity,

$$\lim_{x\to c} f(x) = f(c).$$

Also, because f and g are differentiable at c,

$$\lim_{x\to c} \frac{f(x)-f(c)}{x-c} = f'(c)$$

and

$$\lim_{x\to c} \frac{g(x)-g(c)}{x-c} = g'(c).$$

Consequently,

$$(fg)'(c) = f(c)g'(c) + g(c)f'(c).$$

This proves that fg is differentiable at c and establishes the product rule. □

Problem 54. *Find $f'(x)$ when $f : \mathbb{R} \to \mathbb{R}$ is given by $f(x) = x^n$. Remember this means f' is a function of x and you have to evaluate it at every point of the domain.*

Problem 55. *Suppose that f is differentiable at x. Prove that*

$$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x-h)}{2h}.$$

Hint: Add and subtract $f(x)$ in the numerator.

Theorem 17 (Derivative of the Reciprocal)**.** *If g is differentiable at a and $g(a) \neq 0$, then $1/g$ is differentiable at a, and*

$$\left(\frac{1}{g}\right)'(a) = -\frac{g'(a)}{[g(a)]^2}.$$

Proof. Before we even write

$$\frac{\left(\frac{1}{g}\right)(a+h) - \left(\frac{1}{g}\right)(a)}{h},$$

we must be sure that this expression makes sense. In particular, we need to check that $g(a+h) \neq 0$ for sufficiently small h.

Since g is differentiable at a, it is continuous at a. Moreover, $g(a) \neq 0$. By continuity, there exists $\delta > 0$ such that

$$g(a+h) \neq 0 \quad \text{whenever } |h| < \delta.$$

Thus, $1/g$ is well-defined in a neighborhood of a.

We can now compute

$$\begin{aligned}
\left(\frac{1}{g}\right)'(a) &= \lim_{h\to 0} \frac{\left(\frac{1}{g}\right)(a+h) - \left(\frac{1}{g}\right)(a)}{h} \\
&= \lim_{h\to 0} \frac{\frac{1}{g(a+h)} - \frac{1}{g(a)}}{h} \\
&= \lim_{h\to 0} \frac{g(a) - g(a+h)}{h\, g(a)\, g(a+h)} \\
&= \lim_{h\to 0} \left(-\frac{g(a+h) - g(a)}{h} \cdot \frac{1}{g(a)\, g(a+h)} \right) \\
&= -\lim_{h\to 0} \frac{g(a+h) - g(a)}{h} \cdot \lim_{h\to 0} \frac{1}{g(a)\, g(a+h)} \\
&= -g'(a) \cdot \frac{1}{[g(a)]^2} \\
&= -\frac{g'(a)}{[g(a)]^2}.
\end{aligned}$$

Here, we have used the continuity of g at a once again to conclude that

$$\lim_{h\to 0} g(a+h) = g(a).$$

□

Theorem 18 (The Chain Rule)**.** *Suppose g is differentiable at a, and f is differentiable at $g(a)$. Then $f \circ g$ is differentiable at a, and*

$$(f \circ g)'(a) = f'(g(a)) \cdot g'(a).$$

Proof. Define a function ϕ as follows:

$$\phi(h) = \begin{cases} \dfrac{f(g(a+h)) - f(g(a))}{g(a+h) - g(a)}, & \text{if } g(a+h) - g(a) \neq 0, \\ f'(g(a)), & \text{if } g(a+h) - g(a) = 0. \end{cases}$$

We first prove that ϕ is continuous at 0.

Intuitively, this should be clear. When h is small, $g(a+h) - g(a)$ is also small, so if $g(a+h) - g(a) \neq 0$, then $\phi(h)$ should be close to $f'(g(a))$. On the other hand, if $g(a+h) - g(a) = 0$, then $\phi(h) = f'(g(a))$ by definition.

We now give a rigorous proof of this observation.

Since f is differentiable at $g(a)$, we know that

$$\lim_{k\to 0} \frac{f(g(a)+k) - f(g(a))}{k} = f'(g(a)).$$

Thus, given $\epsilon > 0$, there exists $\delta' > 0$ such that whenever $0 < |k| < \delta'$, we have

$$\left| \frac{f(g(a)+k) - f(g(a))}{k} - f'(g(a)) \right| < \epsilon.$$

Now, since g is differentiable at a, it is continuous at a. Therefore, there exists $\delta > 0$ such that whenever $|h| < \delta$, we have

$$|g(a+h) - g(a)| < \delta'.$$

Consider any h with $|h| < \delta$.

Case 1: Suppose $g(a+h) - g(a) \neq 0$.

Let

$$k = g(a+h) - g(a).$$

Then $0 < |k| < \delta'$, and

$$\phi(h) = \frac{f(g(a+h)) - f(g(a))}{g(a+h) - g(a)} = \frac{f(g(a)+k) - f(g(a))}{k}.$$

By our choice of δ', it follows that

$$|\phi(h) - f'(g(a))| < \epsilon.$$

Case 2: Suppose $g(a+h) - g(a) = 0$.

In this case, by the definition of ϕ,

$$\phi(h) = f'(g(a)).$$

Hence,

$$|\phi(h) - f'(g(a))| = 0 < \epsilon.$$

Thus, in both cases, whenever $|h| < \delta$, we have

$$|\phi(h) - f'(g(a))| < \epsilon.$$

We have therefore proved that

$$\lim_{h \to 0} \phi(h) = f'(g(a)).$$

Since $\phi(0) = f'(g(a))$, this shows that ϕ is continuous at 0.

Now, for every sufficiently small $h \neq 0$, we have

$$\frac{f(g(a+h)) - f(g(a))}{h} = \phi(h) \cdot \frac{g(a+h) - g(a)}{h}.$$

Notice that this equality holds even when $g(a+h) - g(a) = 0$, because in that case both sides are equal to 0.

Taking limits as $h \to 0$, we obtain

$$\begin{aligned}
(f \circ g)'(a) &= \lim_{h \to 0} \frac{f(g(a+h)) - f(g(a))}{h} \\
&= \lim_{h \to 0} \left(\phi(h) \cdot \frac{g(a+h) - g(a)}{h} \right) \\
&= \left(\lim_{h \to 0} \phi(h) \right) \left(\lim_{h \to 0} \frac{g(a+h) - g(a)}{h} \right) \\
&= f'(g(a)) \cdot g'(a).
\end{aligned}$$

This completes the proof. □

8.2 Critical points, local maxima-minimal and the Mean value theorem

Discussion: What would be a sequential definition of continuity?

Definition 26. *Let $f : (a, b) \to \mathbb{R}$ be a continuous function. We say that f attains a local maximum (or minimum) at $c \in (a, b)$ if there is some $\epsilon > 0$ such that $f(c) \geq f(x)$ (or $f(c) \leq f(x)$) for every $x \in (c - \epsilon, c + \epsilon)$.*

Why is a *local* maxima/minima not exactly the maxima/minima of the whole function?

Theorem 19 (Extremal Points are Critical Points)**.** *Suppose $f : (a, b) \to \mathbb{R}$ is differentiable and attains a local maximum at some point $c \in (a, b)$. That is, suppose there exists $\epsilon > 0$ such that for all $x \in (c-\epsilon, c+\epsilon)$, we have $f(x) \leq f(c)$. Then $f'(c) = 0$.*

Definition 27. *A point c where $f'(c) = 0$ is called a critical point of f.*

Proof. Consider the sequence $x_n = c - \frac{1}{n}$. For all sufficiently large n, we have $x_n \in (a, c)$ and $x_n \in (c - \epsilon, c + \epsilon)$. Since f is differentiable at c, we have

$$f'(c) = \lim_{n\to\infty} \frac{f(x_n) - f(c)}{x_n - c}.$$

Since f attains a local maximum at c, we have $f(x_n)-f(c) \leq 0$ for all sufficiently large n. Moreover, $x_n - c = -\frac{1}{n} < 0$. Thus,

$$\frac{f(x_n) - f(c)}{x_n - c} \geq 0.$$

Taking the limit, we obtain $f'(c) \geq 0$.

Similarly, consider the sequence $x'_n = c + \frac{1}{n}$. For all sufficiently large n, we have $x'_n \in (a, b)$ and $f(x'_n) \leq f(c)$. Since $x'_n - c = \frac{1}{n} > 0$, we obtain

$$\frac{f(x'_n) - f(c)}{x'_n - c} \leq 0.$$

Taking the limit gives $f'(c) \leq 0$.

Therefore, $f'(c) \geq 0$ and $f'(c) \leq 0$, which implies

$$f'(c) = 0.$$

□

Theorem 20 (Rolle's Theorem)**.** *Suppose $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b). Suppose also that $f(a) = f(b) = 0$. Then there exists a point $c \in (a, b)$ such that $f'(c) = 0$, i.e., a critical point of f.*

Proof. Since f is continuous on the closed interval $[a, b]$, it attains both a maximum and a minimum by the Extreme Value Theorem.

We consider the following three cases.

Case 1: The maximum of f is strictly bigger than 0. Then the maximum of f occurs at some point $c \in (a, b)$, since $f(a) = f(b) = 0$. By the previous theorem (Extremal Points are Critical Points), we have $f'(c) = 0$.

Case 2: The minimum of f is strictly less than 0. Then the minimum of f occurs at some point $c \in (a, b)$, since $f(a) = f(b) = 0$. Again, by the previous theorem, we have $f'(c) = 0$.

Case 3: The maximum and minimum of f are both 0. In this case, f is identically 0 on $[a, b]$. Hence $f'(c) = 0$ for every $c \in (a, b)$.

In all three cases, there exists $c \in (a, b)$ such that $f'(c) = 0$. This concludes the proof. □

Discussion: What if in the above theorem I replaced $f(a) = f(b) = 0$ with $f(a) = f(b) = l$?

Theorem 21 (Mean Value Theorem)**.** *Suppose $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b). Then there exists a point $c \in (a, b)$ such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

That is, the instantaneous rate of change at c is equal to the average rate of change between a and b.

Let us first draw and test our intuition.

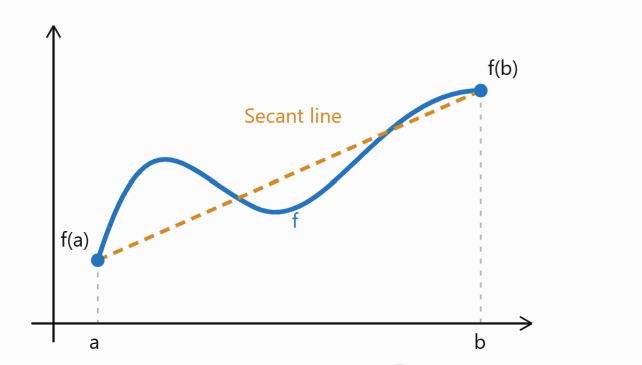


Figure 4: Geometric interpretation of the Mean Value Theorem

We now give a proof:

Proof. Consider the function $g : [a, b] \to \mathbb{R}$ defined by

$$g(x) = f(x) - \frac{f(b) - f(a)}{b - a}(x - a).$$

Since f is continuous on $[a,b]$ and differentiable on (a,b), the same is true for g.

Notice that

$$g(a) = f(a)$$

and

$$g(b) = f(b) - \frac{f(b)-f(a)}{b-a}(b-a) = f(a).$$

Thus, $g(a) = g(b)$.

By Rolle's Theorem, there exists some $c \in (a,b)$ such that $g'(c) = 0$.

Now, differentiating g, we obtain

$$g'(x) = f'(x) - \frac{f(b)-f(a)}{b-a}.$$

Therefore,

$$0 = g'(c) = f'(c) - \frac{f(b)-f(a)}{b-a},$$

which implies

$$f'(c) = \frac{f(b)-f(a)}{b-a}.$$

This completes the proof. □

Corollary 3. *Let $f : [a,b] \to \mathbb{R}$ be continuous on $[a,b]$ and differentiable on (a,b). Suppose that $f'(x) = m$ for every $x \in (a,b)$, where m is a constant. Then there exists $c \in \mathbb{R}$ such that*

$$f(x) = mx + c$$

for all $x \in [a,b]$.

Proof. Define $g : [a,b] \to \mathbb{R}$ by

$$g(x) = f(x) - mx.$$

We claim that g is constant. Indeed, for every $x \in (a,b)$,

$$g'(x) = f'(x) - m = 0.$$

Now let $x, y \in [a,b]$ with $x < y$. By the Mean Value Theorem, there exists $z \in (x,y)$ such that

$$\frac{g(y)-g(x)}{y-x} = g'(z) = 0.$$

Consequently, $g(y) = g(x)$. Since x and y were arbitrary, g must be constant on $[a,b]$.

Thus, there exists $c \in \mathbb{R}$ such that

$$f(x) - mx = c,$$

or equivalently,

$$f(x) = mx + c.$$

This proves the result. □

Problem 56. *Assuming the trignometric identity of addition of* $\sin x$*, find its derivative.*

After deriving the derivative above from first principles, can you find the derivative of $\sin(\sin^2 x)$*?*

Now that we have discussed Mean value theorem, we begin proving and understanding some statements that are obvious intuitively. Our job is now to make them mathematically precise.

Definition 28. *Let* f *be a function defined on an interval* I*. We say that* f *is* ***increasing*** *on* I *if* $f(a) < f(b)$ *whenever* $a, b \in I$ *with* $a < b$*.*

Similarly, we say that f *is* ***decreasing*** *on* I *if* $f(a) > f(b)$ *whenever* $a, b \in I$ *with* $a < b$*.*

Corollary 4. *Let* f *be continuous on an interval* I *and differentiable at every interior point of* I*. If* $f'(x) > 0$ *for every interior point* $x \in I$*, then* f *is increasing on* I*. Similarly, if* $f'(x) < 0$ *for every interior point* $x \in I$*, then* f *is decreasing on* I*.*

Proof. Suppose first that $f'(x) > 0$ for every interior point $x \in I$. Let $a, b \in I$ with $a < b$. By the Mean Value Theorem, there exists $c \in (a, b)$ such that

$$\frac{f(b) - f(a)}{b - a} = f'(c) > 0.$$

Since $b - a > 0$, we conclude that $f(b) > f(a)$. Thus f is increasing on I.

Similarly, if $f'(x) < 0$ for every interior point $x \in I$, the Mean Value Theorem gives

$$\frac{f(b) - f(a)}{b - a} = f'(c) < 0.$$

Since $b - a > 0$, we have $f(b) < f(a)$. Hence f is decreasing on I. □

But, is it always true that if you have an increasing function, its derivative will be strictly bigger than 0? Think about x^3 at $x = 0$. Our intuition can defy us, that is why definitions and being careful with every statement in math is important.

Theorem 22 (Second Derivative Test)**.** *Let* f *be twice differentiable at* a *and suppose that* $f'(a) = 0$*.*

1. *If* $f''(a) > 0$*, then* f *has a local minimum at* a*.*
2. *If* $f''(a) < 0$*, then* f *has a local maximum at* a*.*

Proof. By the definition of the second derivative,

$$f''(a) = \lim_{h \to 0} \frac{f'(a + h) - f'(a)}{h}.$$

Since $f'(a) = 0$, this becomes

$$f''(a) = \lim_{h \to 0} \frac{f'(a+h)}{h}.$$

Suppose first that $f''(a) > 0$. By the definition of a limit, there exists $\delta > 0$ such that

$$\frac{f'(a+h)}{h} > 0$$

whenever $0 < |h| < \delta$.

In particular, if $0 < h < \delta$, then $f'(a+h) > 0$, while if $-\delta < h < 0$, then $f'(a+h) < 0$.

Thus, f is decreasing on $(a-\delta, a)$ and increasing on $(a, a+\delta)$. Consequently,

$$f(x) \geq f(a)$$

for all $x \in (a-\delta, a+\delta)$. Hence f has a local minimum at a.

Similarly, suppose that $f''(a) < 0$. Then there exists $\delta > 0$ such that

$$\frac{f'(a+h)}{h} < 0$$

whenever $0 < |h| < \delta$.

Therefore, $f'(a+h) < 0$ for sufficiently small $h > 0$, and $f'(a+h) > 0$ for sufficiently small $h < 0$. It follows that f is increasing to the left of a and decreasing to the right of a. Thus,

$$f(x) \leq f(a)$$

for all x sufficiently close to a, proving that f has a local maximum at a. □

Theorem 23. *Suppose that $f''(a)$ exists. If f has a local minimum at a, then $f''(a) \geq 0$. Similarly, if f has a local maximum at a, then $f''(a) \leq 0$.*

Proof. Suppose that f has a local minimum at a. We want to show that $f''(a) \geq 0$.

Assume, towards a contradiction, that $f''(a) < 0$. Since f has a local minimum at a and is differentiable there, we have $f'(a) = 0$.

By the Second Derivative Test, $f''(a) < 0$ implies that f also has a local maximum at a. Thus, f has both a local minimum and a local maximum at a.

Consequently, f must be constant on some open interval containing a. In particular, $f''(a) = 0$, which contradicts our assumption that $f''(a) < 0$.

Therefore, $f''(a) \geq 0$.

The proof for the case where f has a local maximum at a is similar. In that case, assuming $f''(a) > 0$ would imply that f also has a local minimum at a, leading to the same contradiction. Hence $f''(a) \leq 0$. □

Recall when we stated the mean value theorem, we were dealing with a function on the closed interval $[a, b]$, that was differentiable on (a, b) and continuous on a. What if we are now thinking about the boundary point a and whether it can be assigned a right hand derivative. The following theorem deals with this case and more, because it is not necessarily about a boundary point. The theorem is trying to answer: How can we relate a secant slope involving a to a derivative at a point other than a?

Theorem 24. *Suppose that f is continuous at a and that $f'(x)$ exists for all x in some open interval containing a, except perhaps at $x = a$. Suppose, moreover, that*

$$\lim_{x \to a} f'(x)$$

exists. Then $f'(a)$ also exists, and

$$f'(a) = \lim_{x \to a} f'(x).$$

Proof. We want to show that the limit

$$\lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

exists and equals $\lim_{x \to a} f'(x)$.

Let $h > 0$ be sufficiently small. Since f is continuous at a and differentiable at every point of $(a, a + h)$, it is continuous on $[a, a + h]$ and differentiable on $(a, a + h)$.

By the Mean Value Theorem, there exists a point $c_h \in (a, a + h)$ such that

$$\frac{f(a+h) - f(a)}{h} = f'(c_h).$$

Notice that

$$a < c_h < a + h.$$

As $h \to 0^+$, the Squeeze Theorem implies that $c_h \to a$. Since $\lim_{x \to a} f'(x)$ exists, we conclude that

$$\lim_{h \to 0^+} \frac{f(a+h) - f(a)}{h} = \lim_{h \to 0^+} f'(c_h) = \lim_{x \to a} f'(x).$$

Similarly, for sufficiently small $h < 0$, the Mean Value Theorem applied to $[a + h, a]$ gives a point $c_h \in (a + h, a)$ such that

$$\frac{f(a+h) - f(a)}{h} = f'(c_h).$$

Again, $c_h \to a$ as $h \to 0^-$, and therefore

$$\lim_{h \to 0^-} \frac{f(a+h) - f(a)}{h} = \lim_{x \to a} f'(x).$$

Thus, both one-sided limits exist and are equal. Consequently,

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} = \lim_{x \to a} f'(x).$$

This completes the proof. □

Example 6. *The preceding theorem tells us that if f is continuous at a and $\lim_{x \to a} f'(x)$ exists, then f is differentiable at a and*

$$f'(a) = \lim_{x \to a} f'(x).$$

However, the converse is not necessarily true. In other words, even if f is differentiable at a, the limit $\lim_{x \to a} f'(x)$ need not exist.

To see this, consider the function

$$f(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

We first show that f is differentiable at 0. By the definition of the derivative,

$$f'(0) = \lim_{h \to 0} \frac{f(h) - f(0)}{h} = \lim_{h \to 0} h \sin(1/h) = 0,$$

where the last equality follows from the Squeeze Theorem, since

$$-|h| \leq h \sin(1/h) \leq |h|.$$

Thus, $f'(0)$ exists and equals 0.

On the other hand, for $x \neq 0$, the product and chain rules give

$$f'(x) = 2x \sin(1/x) - \cos(1/x).$$

Notice that

$$\lim_{x \to 0} 2x \sin(1/x) = 0,$$

but $\cos(1/x)$ oscillates between -1 and 1 as x approaches 0 and has no limit. Consequently,

$$\lim_{x \to 0} f'(x)$$

does not exist.

Therefore, f is differentiable at 0, but its derivative f' is not continuous at 0.

This example shows that differentiability of a function does not imply continuity of its derivative. In particular, the converse of the preceding theorem is false.

The previous theorem and example were showing us that derivative might exist at a point, but it need not be continuous!

Theorem 25 (Cauchy Mean Value Theorem)**.** *Let $f, g : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Then there exists some $c \in (a, b)$ such that*

$$[f(b) - f(a)]g'(c) = [g(b) - g(a)]f'(c).$$

In particular, if $g(b) \neq g(a)$ and $g'(x) \neq 0$ for every $x \in (a, b)$, then

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}.$$

Remark 4. *Notice that if $g(x) = x$, then $g'(x) = 1$, and the Cauchy Mean Value Theorem reduces to the usual Mean Value Theorem.*

One might try to prove the result by applying the Mean Value Theorem separately to f and g. However, this would only give points $c_1, c_2 \in (a, b)$ such that

$$f(b) - f(a) = f'(c_1)(b - a)$$

and

$$g(b) - g(a) = g'(c_2)(b - a).$$

Consequently,

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c_1)}{g'(c_2)},$$

provided the denominator is nonzero.

The difficulty is that there is no reason for c_1 and c_2 to be equal. The Cauchy Mean Value Theorem tells us that we can find a single point c at which the desired equality holds.

Proof. Consider the auxiliary function $h : [a, b] \to \mathbb{R}$ defined by

$$h(x) = f(x)[g(b) - g(a)] - g(x)[f(b) - f(a)].$$

Since f and g are continuous on $[a, b]$ and differentiable on (a, b), so is h.

Moreover,

$$\begin{aligned} h(a) &= f(a)[g(b) - g(a)] - g(a)[f(b) - f(a)] \\ &= f(a)g(b) - g(a)f(b), \end{aligned}$$

and similarly,

$$\begin{aligned} h(b) &= f(b)[g(b) - g(a)] - g(b)[f(b) - f(a)] \\ &= f(a)g(b) - g(a)f(b). \end{aligned}$$

Thus, $h(a) = h(b)$.

By Rolle's Theorem, there exists some $c \in (a, b)$ such that $h'(c) = 0$. But

$$h'(x) = f'(x)[g(b) - g(a)] - g'(x)[f(b) - f(a)].$$

Therefore,

$$0 = f'(c)[g(b) - g(a)] - g'(c)[f(b) - f(a)],$$

which gives

$$[f(b) - f(a)]g'(c) = [g(b) - g(a)]f'(c).$$

This proves the Cauchy Mean Value Theorem. □

Theorem 26 (L'Hôpital's Rule: The 0/0 Case)**.** *Let f and g be functions defined on a punctured neighborhood of a, where $a \in \mathbb{R}$. Suppose that*

$$\lim_{x \to a} f(x) = 0 \qquad \textit{and} \qquad \lim_{x \to a} g(x) = 0.$$

Assume that f and g are differentiable for all x sufficiently close to a, with the possible exception of $x = a$, and that $g'(x) \neq 0$ for all such $x \neq a$.

If

$$\lim_{x \to a} \frac{f'(x)}{g'(x)} = L$$

exists, where $L \in \mathbb{R}$, then

$$\lim_{x \to a} \frac{f(x)}{g(x)} = L.$$

Proof. The main idea is to use the Cauchy Mean Value Theorem to relate the quotient $f(x)/g(x)$ to the quotient of derivatives $f'(x)/g'(x)$.

First, notice that the values of f and g at a do not affect the limits we are trying to compute. We may therefore define

$$f(a) = g(a) = 0.$$

Since

$$\lim_{x \to a} f(x) = \lim_{x \to a} g(x) = 0,$$

both functions are now continuous at a.

We first consider $x > a$, sufficiently close to a. Since $g'(t) \neq 0$ for all $t \in (a, x)$, the Mean Value Theorem implies that $g(x) \neq g(a) = 0$. Indeed, if $g(x) = g(a)$, Rolle's Theorem would give a point $t \in (a, x)$ such that $g'(t) = 0$, a contradiction.

Now apply the Cauchy Mean Value Theorem to f and g on $[a, x]$. There exists a point $c_x \in (a, x)$ such that

$$[f(x) - f(a)]g'(c_x) = [g(x) - g(a)]f'(c_x).$$

Using $f(a) = g(a) = 0$, we obtain

$$f(x)g'(c_x) = g(x)f'(c_x).$$

Since $g(x) \neq 0$ and $g'(c_x) \neq 0$, we may divide to get

$$\frac{f(x)}{g(x)} = \frac{f'(c_x)}{g'(c_x)}.$$

The crucial observation is that c_x lies between a and x. Thus,

$$a < c_x < x,$$

and as $x \to a^+$, the Squeeze Theorem gives $c_x \to a$.

By our assumption,

$$\lim_{t\to a}\frac{f'(t)}{g'(t)}=L.$$

Consequently,

$$\lim_{x\to a^+}\frac{f(x)}{g(x)}=\lim_{x\to a^+}\frac{f'(c_x)}{g'(c_x)}=L.$$

The same argument applies when $x<a$, using the interval $[x,a]$. It follows that

$$\lim_{x\to a^-}\frac{f(x)}{g(x)}=L.$$

Since both one-sided limits exist and are equal, we conclude that

$$\boxed{\lim_{x\to a}\frac{f(x)}{g(x)}=\lim_{x\to a}\frac{f'(x)}{g'(x)}.}$$

□

9 Inverse functions

In the Problem Set 7.3, you learnt to work with inverse functions and show that if a function is continuous, so is its inverse. What about differentiability? For instance, lets look at the following examples near $x=0$.

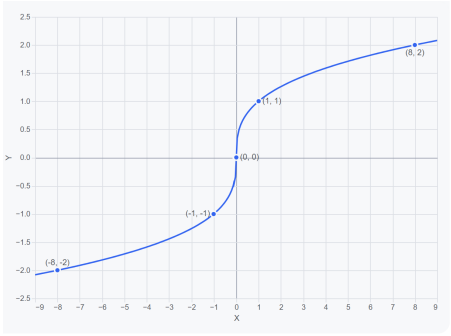


Figure 5: Caption

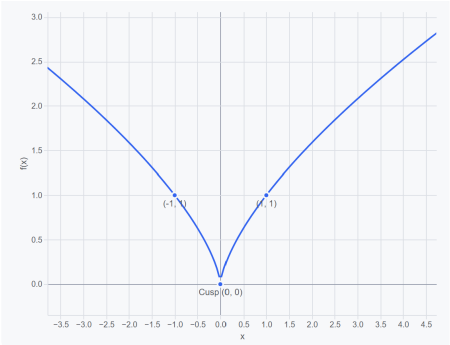


Figure 6: Graph of $y = x^{2/3}$

In the first example, imagine walking along the curve from left to right. On the far left, the path is relatively flat. As you approach $x = 0$, the hill gets steeper and steeper. It almost looks like in the first example, near $x = 0$, the slope is parallel to the y-axis. What is the value of the slope of y-axis?

In the second case, you see another phenomena, that you would not expect from a differentiable function: a sharp edge.

Theorem 27. *Let f be a continuous injective function defined on an interval I. Suppose that f is differentiable at $f^{-1}(b)$ and that $f'(f^{-1}(b)) \neq 0$. Then f^{-1} is differentiable at b, and*

$$(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))}.$$

Proof. Let $a = f^{-1}(b)$, so that $f(a) = b$. By the definition of the derivative, we wish to evaluate

$$\lim_{h \to 0} \frac{f^{-1}(b+h) - f^{-1}(b)}{h}.$$

For every h such that $b + h$ belongs to the domain of f^{-1}, we can write

$$b + h = f(a + k)$$

for a unique k. In particular,

$$k = f^{-1}(b + h) - f^{-1}(b).$$

Since f^{-1} is continuous at b, we know that $k \to 0$ as $h \to 0$. Moreover, since f is injective, $h \neq 0$ implies $k \neq 0$.

We can therefore rewrite the difference quotient as

$$\begin{aligned}\frac{f^{-1}(b+h)-f^{-1}(b)}{h} &= \frac{f^{-1}(f(a+k))-a}{f(a+k)-f(a)}\\ &= \frac{k}{f(a+k)-f(a)}\\ &= \frac{1}{\dfrac{f(a+k)-f(a)}{k}}.\end{aligned}$$

As $h \to 0$, we have $k \to 0$. Since f is differentiable at a and $f'(a) \neq 0$, it follows that

$$\begin{aligned}(f^{-1})'(b) &= \lim_{h\to 0}\frac{f^{-1}(b+h)-f^{-1}(b)}{h}\\ &= \frac{1}{\lim\limits_{k\to 0}\dfrac{f(a+k)-f(a)}{k}}\\ &= \frac{1}{f'(a)} = \frac{1}{f'(f^{-1}(b))}.\end{aligned}$$

Hence f^{-1} is differentiable at b, with the claimed derivative. □

9.0.1 Problem set 2

Problem 57. *Let b be odd. For which rational numbers $\frac{a}{b}$, where a, b are coprime positive, does the derivative at $x = 1$ of the function $f(x) = (x-1)^{a/b}$ exist? In the case when derivative exists, calculate its value at all x. (Hint: Think of $x^{a/b}$ as a composition of two functions.)*

Problem 58. *Suppose $f(0) = 0$ and f' is increasing. Prove that the function $g(x) = \frac{f(x)}{x}$ is increasing on $(0, \infty)$. Take the example of $f(x) = 1 + x^2$ and explain why the hypothesis of $f(x) = 0$ is important in the above problem.*

Problem 59. *Use derivatives to prove that for every integer $n \geq 1$,*

$$(1+x)^n > 1 + nx$$

whenever $-1 < x < 0$ or $x > 0$.

Notice that equality holds when $x = 0$.

Problem 60. *What is wrong with the following application of L'Hôpital's Rule?*

$$\lim_{x\to 1}\frac{x^3+x-2}{x^2-3x+2} = \lim_{x\to 1}\frac{3x^2+1}{2x-3} = \lim_{x\to 1}\frac{6x}{2} = 3.$$

Show that the limit is -4.

Problem 61. *Find $f'(0)$ if*

$$f(x) = \begin{cases} \dfrac{g(x)}{x}, & x \neq 0, \\ 0, & x = 0, \end{cases}$$

where $g(0) = g'(0) = 0$ and $g''(0) = 17$.

Problem 62. *Prove that*

$$\frac{1}{9} < \sqrt{66} - 8 < \frac{1}{8}.$$

Your proof should not involve computing a decimal expansion of $\sqrt{66}$.

10 Integrals

Example 7. *Consider a right triangle with base b and height h. Suppose its hypotenuse lies on the line*

$$y = mx,$$

where $m > 0$, and one vertex is at the origin.

For each $x > 0$, let $A(x)$ denote the area of the right triangle whose base is $[0, x]$ and whose height is mx.

Recall that the area of a right triangle is half the product of its base and height. Thus,

$$A(x) = \frac{1}{2}x(mx) = \frac{m}{2}x^2.$$

Differentiating, we obtain

$$A'(x) = mx.$$

Notice that mx is precisely the height of the triangle at the point x. In other words,

$$\boxed{A'(x) = f(x), \qquad f(x) = mx.}$$

This suggests an interesting relationship between area and differentiation: the derivative of the area accumulated up to a point x is the height of the region at that point.

More generally, suppose we have a nonnegative continuous function f and let $A(x)$ denote the area under its graph from 0 to x.

Can we expect that

$$A'(x) = f(x)?$$

We will make this observation precise in our study of integration.

Problem 63. *Above I took the example of a constant function to show that area under it when differentiated gives the constant function back. Convince yourself with another example, whose area would be easier to calculate.*

Definition 29. *Let $a < b$. A **partition** of the interval $[a, b]$ is a finite set of points in $[a, b]$ containing both a and b.*

We write a partition as $P = \{t_0, t_1, \ldots, t_n\}$, where

$$a = t_0 < t_1 < \cdots < t_{n-1} < t_n = b.$$

We will always assume that the points of a partition are arranged in this order.

Definition 30. *Let f be a bounded function on $[a, b]$, and let $P = \{t_0, t_1, \ldots, t_n\}$ be a partition of $[a, b]$. For each $i = 1, \ldots, n$, define*

$$m_i = \inf\{f(x) : t_{i-1} \le x \le t_i\},$$
$$M_i = \sup\{f(x) : t_{i-1} \le x \le t_i\}.$$

*The **lower sum** of f with respect to P, denoted by $L(f, P)$, is defined by*

$$L(f, P) = \sum_{i=1}^{n} m_i(t_i - t_{i-1}).$$

*Similarly, the **upper sum** of f with respect to P, denoted by $U(f, P)$, is defined by*

$$U(f, P) = \sum_{i=1}^{n} M_i(t_i - t_{i-1}).$$

Remark 5. *Let f be a bounded function on $[a, b]$. For any partition $P = \{t_0, t_1, \ldots, t_n\}$ of $[a, b]$, we have*

$$L(f, P) \le U(f, P).$$

Indeed, recall that

$$L(f, P) = \sum_{i=1}^{n} m_i(t_i - t_{i-1})$$

and

$$U(f, P) = \sum_{i=1}^{n} M_i(t_i - t_{i-1}).$$

Since $m_i \le M_i$ and $t_i - t_{i-1} > 0$ for every i, we obtain

$$m_i(t_i - t_{i-1}) \le M_i(t_i - t_{i-1}).$$

Summing these inequalities over $i = 1, \ldots, n$ gives

$$\boxed{L(f, P) \le U(f, P).}$$

Lemma 2. *Let f be a bounded function on $[a, b]$, and let P and Q be two partitions of $[a, b]$. Suppose that Q is a refinement of P, i.e., every point of P also belongs to Q. Then*

$$L(f, P) \le L(f, Q)$$

and

$$U(f, P) \ge U(f, Q).$$

Proof. We first consider the case where Q is obtained from P by adding exactly one point. Write

$$P = \{t_0, t_1, \ldots, t_n\}$$

and

$$Q = \{t_0, \ldots, t_{k-1}, u, t_k, \ldots, t_n\},$$

where

$$a = t_0 < \cdots < t_{k-1} < u < t_k < \cdots < t_n = b.$$

Let

$$m' = \inf\{f(x) : t_{k-1} \le x \le u\},$$
$$m'' = \inf\{f(x) : u \le x \le t_k\}.$$

Recall that

$$m_k = \inf\{f(x) : t_{k-1} \le x \le t_k\}.$$

Since both $[t_{k-1}, u]$ and $[u, t_k]$ are contained in $[t_{k-1}, t_k]$, we have

$$m_k \le m', \qquad m_k \le m''.$$

Consequently,

$$\begin{aligned} m_k(t_k - t_{k-1}) &= m_k(u - t_{k-1}) + m_k(t_k - u) \\ &\le m'(u - t_{k-1}) + m''(t_k - u). \end{aligned}$$

Now observe that

$$L(f, P) = \sum_{i=1}^{n} m_i(t_i - t_{i-1}),$$

whereas

$$\begin{aligned} L(f, Q) = &\sum_{i=1}^{k-1} m_i(t_i - t_{i-1}) \\ &+ m'(u - t_{k-1}) + m''(t_k - u) \\ &+ \sum_{i=k+1}^{n} m_i(t_i - t_{i-1}). \end{aligned}$$

The only difference between these two sums comes from the interval $[t_{k-1}, t_k]$. By the inequality above, we conclude that

$$L(f, P) \le L(f, Q).$$

The argument for upper sums is analogous. Define

$$M' = \sup\{f(x) : t_{k-1} \le x \le u\},$$
$$M'' = \sup\{f(x) : u \le x \le t_k\}.$$

Since

$$M_k \ge M', \qquad M_k \ge M'',$$

we obtain

$$\begin{aligned} M_k(t_k - t_{k-1}) &= M_k(u - t_{k-1}) + M_k(t_k - u) \\ &\geq M'(u - t_{k-1}) + M''(t_k - u). \end{aligned}$$

Thus,

$$U(f, P) \geq U(f, Q).$$

We now consider the general case. Since Q is a refinement of P, it can be obtained by adding finitely many points to P, one at a time. Hence there exists a sequence of partitions

$$P = P_1, P_2, \ldots, P_r = Q$$

such that P_{j+1} is obtained from P_j by adding exactly one point.

Applying the special case repeatedly gives

$$\begin{aligned} L(f, P) = L(f, P_1) &\leq L(f, P_2) \leq \cdots \\ &\leq L(f, P_r) = L(f, Q), \end{aligned}$$

and similarly,

$$\begin{aligned} U(f, P) = U(f, P_1) &\geq U(f, P_2) \geq \cdots \\ &\geq U(f, P_r) = U(f, Q). \end{aligned}$$

This proves both inequalities. □

Theorem 28. *Let f be a bounded function on $[a, b]$, and let P_1 and P_2 be any two partitions of $[a, b]$. Then*

$$L(f, P_1) \leq U(f, P_2).$$

Proof. Let $P = P_1 \cup P_2$ be the partition consisting of all points belonging to either P_1 or P_2. Then P is a refinement of both P_1 and P_2.

By the preceding lemma, we have

$$L(f, P_1) \leq L(f, P)$$

and

$$U(f, P) \leq U(f, P_2).$$

Moreover, we know that

$$L(f, P) \leq U(f, P).$$

Combining these inequalities, we obtain

$$L(f, P_1) \leq L(f, P) \leq U(f, P) \leq U(f, P_2),$$

which proves the theorem. □

Remark 6. *By the preceding theorem, for any two partitions P and P' of $[a,b]$, we have*

$$L(f,P) \le U(f,P').$$

Since this holds for every partition P, we obtain

$$\sup\{L(f,P) : P \text{ is a partition of } [a,b]\} \le U(f,P').$$

As P' was arbitrary, the supremum of all lower sums is a lower bound for the set of all upper sums. Consequently,

$$\boxed{\sup_P L(f,P) \le \inf_P U(f,P).}$$

Here, the supremum and infimum are taken over all partitions of $[a,b]$.

Problem 64 (Discussion)**.** *Why do the following quantities exist as finite real numbers?*

$$\sup\{L(f,P) : P \text{ is a partition of } [a,b]\}$$

and

$$\inf\{U(f,P) : P \text{ is a partition of } [a,b]\}.$$

Recall that f is assumed to be bounded on $[a,b]$. Which property of the real numbers guarantees the existence of these quantities?

Example 8. *We now give an example to show that the supremum of the lower sums need not equal the infimum of the upper sums.*

Consider the function $f : [a,b] \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} 0, & x \text{ is irrational,} \\ 1, & x \text{ is rational.} \end{cases}$$

Let $P = \{t_0, t_1, \ldots, t_n\}$ be any partition of $[a,b]$. Since every interval $[t_{i-1}, t_i]$ contains both rational and irrational numbers, we have

$$m_i = 0, \qquad M_i = 1$$

for every $i = 1, \ldots, n$.

Consequently,

$$L(f,P) = \sum_{i=1}^{n} 0 \cdot (t_i - t_{i-1}) = 0$$

and

$$U(f,P) = \sum_{i=1}^{n} (t_i - t_{i-1}) = b - a.$$

Since these equalities hold for every partition P, we conclude that

$$\boxed{\sup_P L(f,P) = 0 < b - a = \inf_P U(f,P).}$$

Thus, the supremum of the lower sums and the infimum of the upper sums need not coincide.

Definition 31. *Let f be a bounded function on $[a,b]$. We say that f is **integrable** on $[a,b]$ if*

$$\sup\{L(f,P) : P \text{ is a partition of } [a,b]\} = \inf\{U(f,P') : P' \text{ is a partition of } [a,b]\}.$$

*When this equality holds, the common value is called the **integral** of f over $[a,b]$, and is denoted by*

$$\int_a^b f(x)\,dx.$$

*The symbol $\int$ is called the **integral sign**. The numbers a and b are called the **lower limit** and **upper limit of integration**, respectively.*

If $f(x) \geq 0$ for every $x \in [a,b]$, then $\int_a^b f(x)\,dx$ represents the area of the region under the graph of f between $x = a$ and $x = b$.

The above definition looks rather complicated to check by hands. To verify it is true we have go through every possible partition to check its infimum and superemum. It would be better if we could find a slightly easier to parse criteria.

Theorem 29. *Let f be a bounded function on $[a,b]$. Then f is integrable on $[a,b]$ if and only if for every $\varepsilon > 0$, there exists a partition P of $[a,b]$ such that*

$$U(f,P) - L(f,P) < \varepsilon.$$

Proof. Suppose first that for every $\varepsilon > 0$, there exists a partition P such that

$$U(f,P) - L(f,P) < \varepsilon.$$

By the definitions of supremum and infimum, we have

$$\inf_{P'} U(f,P') \leq U(f,P), \qquad \sup_{P'} L(f,P') \geq L(f,P).$$

Consequently,

$$0 \leq \inf_{P'} U(f,P') - \sup_{P'} L(f,P') \leq U(f,P) - L(f,P) < \varepsilon.$$

Since this holds for every $\varepsilon > 0$, we conclude that

$$\inf_{P'} U(f,P') = \sup_{P'} L(f,P'),$$

so f is integrable.

Conversely, suppose that f is integrable, and write

$$I = \sup_P L(f,P) = \inf_P U(f,P).$$

Given $\varepsilon > 0$, by the definitions of supremum and infimum, there exist partitions P_1 and P_2 such that

$$I - L(f,P_1) < \frac{\varepsilon}{2}, \qquad U(f,P_2) - I < \frac{\varepsilon}{2}.$$

Let $P = P_1 \cup P_2$ be their common refinement. By the refinement lemma,

$$L(f, P) \geq L(f, P_1), \qquad U(f, P) \leq U(f, P_2).$$

Therefore,

$$\begin{aligned} U(f, P) - L(f, P) &\leq U(f, P_2) - L(f, P_1) \\ &= \big(U(f, P_2) - I\big) + \big(I - L(f, P_1)\big) \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

This completes the proof. □

So basically, we have converted the information of going through all partitions with finding a partition for every ϵ, given the function is bounded. This lets you have much more control while proving integrability of a function. Let us see this through an example:

Example 9. *Let $f(x) = x^2$ on $[0, b]$, where $b > 0$. We will show that f is integrable by using the criterion established in the previous theorem.*

Given $\varepsilon > 0$, we want to find a partition P such that

$$U(f, P) - L(f, P) < \varepsilon.$$

Consider the partition $P_n = \{t_0, t_1, \ldots, t_n\}$ which divides $[0, b]$ into n equal subintervals, so that

$$t_i = \frac{ib}{n}, \qquad i = 0, 1, \ldots, n.$$

Since $f(x) = x^2$ is increasing on $[0, b]$, we have

$$m_i = t_{i-1}^2, \qquad M_i = t_i^2.$$

Therefore,

$$\begin{aligned} U(f, P_n) - L(f, P_n) &= \sum_{i=1}^{n} (M_i - m_i)(t_i - t_{i-1}) \\ &= \sum_{i=1}^{n} (t_i^2 - t_{i-1}^2)\frac{b}{n} \\ &= \frac{b}{n} \sum_{i=1}^{n} (t_i^2 - t_{i-1}^2). \end{aligned}$$

Notice that the sum is telescoping:

$$\begin{aligned} \sum_{i=1}^{n} (t_i^2 - t_{i-1}^2) &= (t_1^2 - t_0^2) + (t_2^2 - t_1^2) + \cdots + (t_n^2 - t_{n-1}^2) \\ &= t_n^2 - t_0^2 = b^2. \end{aligned}$$

Hence,

$$\boxed{U(f, P_n) - L(f, P_n) = \frac{b^3}{n}.}$$

Now, given any $\varepsilon > 0$, choose an integer n such that

$$n > \frac{b^3}{\varepsilon}.$$

Then

$$U(f, P_n) - L(f, P_n) = \frac{b^3}{n} < \varepsilon.$$

By the integrability criterion, $f(x) = x^2$ is integrable on $[0, b]$.

It makes somewhat sense intuitively that a continuous function on a closed interval should be integrable. (Why not an open interval?) What are somethings that make closed interval nice?

Theorem 30. *Every continuous function $f : [a, b] \to \mathbb{R}$ is uniformly continuous on $[a, b]$.*

Proof. Suppose, for the sake of contradiction, that f is not uniformly continuous on $[a, b]$.

Then there exists $\varepsilon_0 > 0$ such that for every $\delta > 0$, we can find $x, y \in [a, b]$ satisfying

$$|x - y| < \delta \qquad \text{but} \qquad |f(x) - f(y)| \geq \varepsilon_0.$$

In particular, taking $\delta = 1/n$, we can find sequences $\{x_n\}$ and $\{y_n\}$ in $[a, b]$ such that

$$|x_n - y_n| < \frac{1}{n}, \qquad |f(x_n) - f(y_n)| \geq \varepsilon_0$$

for every $n \geq 1$.

Since $\{x_n\}$ is bounded, the Bolzano–Weierstrass theorem tells us that it has a convergent subsequence $\{x_{n_k}\}$. Write

$$x_{n_k} \to c.$$

Since $[a, b]$ is closed, we have $c \in [a, b]$.

Now observe that

$$|y_{n_k} - c| \leq |y_{n_k} - x_{n_k}| + |x_{n_k} - c| < \frac{1}{n_k} + |x_{n_k} - c|.$$

Both terms on the right tend to 0, so

$$y_{n_k} \to c.$$

Since f is continuous at c, we obtain

$$f(x_{n_k}) \to f(c), \qquad f(y_{n_k}) \to f(c).$$

Consequently,

$$|f(x_{n_k}) - f(y_{n_k})| \to 0.$$

But this contradicts

$$|f(x_{n_k}) - f(y_{n_k})| \geq \varepsilon_0$$

for every k.

Therefore, f must be uniformly continuous on $[a, b]$. □

The above theorem lets us prove that every continuous function is integrable.

Theorem 31. *Let $f : [a, b] \to \mathbb{R}$ be continuous. Then f is Riemann integrable on $[a, b]$.*

Proof. Since f is continuous on the closed interval $[a, b]$, it is bounded. Thus, it suffices to show that for every $\varepsilon > 0$, there exists a partition P of $[a, b]$ such that

$$U(f, P) - L(f, P) < \varepsilon.$$

Since f is continuous on the closed interval $[a, b]$, it is uniformly continuous. Hence, given $\varepsilon > 0$, there exists $\delta > 0$ such that whenever $x, y \in [a, b]$ satisfy $|x - y| < \delta$, we have

$$|f(x) - f(y)| < \frac{\varepsilon}{2(b-a)}.$$

Choose a partition

$$P = \{a = t_0 < t_1 < \cdots < t_n = b\}$$

such that $t_i - t_{i-1} < \delta$ for every i.

For each i, let

$$M_i = \sup_{x \in [t_{i-1}, t_i]} f(x), \qquad m_i = \inf_{x \in [t_{i-1}, t_i]} f(x).$$

Since f is continuous on each closed subinterval, it attains its maximum and minimum there. By our choice of δ, we therefore have

$$M_i - m_i < \frac{\varepsilon}{2(b-a)}$$

for every $i = 1, \ldots, n$.

Consequently,

$$\begin{aligned} U(f, P) - L(f, P) &= \sum_{i=1}^{n} (M_i - m_i)(t_i - t_{i-1}) \\ &< \frac{\varepsilon}{2(b-a)} \sum_{i=1}^{n} (t_i - t_{i-1}) \\ &= \frac{\varepsilon}{2(b-a)} (b - a) \\ &= \frac{\varepsilon}{2} < \varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ was arbitrary, f is Riemann integrable on $[a, b]$. □

However, we do not need an integrable function f to be continuous. We will show that if a function f is integrable, its integral is continuous.

Definition 32. *Let f be integrable on $[a,b]$, then we define its integral F by*

$$F(x) := \int_a^x f(y)dy.$$

To check the continuity of the above function, we need to have an understanding of $F(c+h) - F(h)$, which is the point of the theorem below.

Theorem 32. *Let $a < c < b$. A bounded function $f : [a,b] \to \mathbb{R}$ is Riemann integrable on $[a,b]$ if and only if it is Riemann integrable on both $[a,c]$ and $[c,b]$.*

Moreover, whenever f is integrable on $[a,b]$, we have

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx.$$

Proof. First, suppose that f is integrable on $[a,b]$. Given $\varepsilon > 0$, choose a partition P of $[a,b]$ such that

$$U(f,P) - L(f,P) < \varepsilon.$$

By adding c to the partition if necessary, we may assume that c is one of its partition points. Indeed, refining a partition can only decrease the upper sum and increase the lower sum.

Write

$$P = \{a = t_0 < t_1 < \cdots < t_k = c < \cdots < t_n = b\}.$$

Let

$$P' = \{t_0, \ldots, t_k\}, \qquad P'' = \{t_k, \ldots, t_n\}$$

be the corresponding partitions of $[a,c]$ and $[c,b]$.

By the definitions of upper and lower sums,

$$U(f,P) = U(f,P') + U(f,P''),$$

and

$$L(f,P) = L(f,P') + L(f,P'').$$

Consequently,

$$\begin{aligned}[U(f,P') - L(f,P')] + [U(f,P'') - L(f,P'')] \\ = U(f,P) - L(f,P) < \varepsilon.\end{aligned}$$

Since both expressions in brackets are nonnegative, each is smaller than ε. Thus f is integrable on both $[a,c]$ and $[c,b]$.

We now establish the formula for the integrals. For any partition P containing c, the inequalities

$$L(f,P') \le \int_a^c f(x)\,dx \le U(f,P')$$

and

$$L(f, P'') \leq \int_c^b f(x)\,dx \leq U(f, P'')$$

give

$$L(f, P) \leq \int_a^c f(x)\,dx + \int_c^b f(x)\,dx \leq U(f, P).$$

On the other hand,

$$L(f, P) \leq \int_a^b f(x)\,dx \leq U(f, P).$$

Since we can choose partitions for which $U(f, P) - L(f, P)$ is arbitrarily small, the two quantities must be equal. Therefore,

$$\boxed{\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx.}$$

Conversely, suppose that f is integrable on $[a, c]$ and $[c, b]$.

Given $\varepsilon > 0$, choose partitions P' of $[a, c]$ and P'' of $[c, b]$ such that

$$U(f, P') - L(f, P') < \frac{\varepsilon}{2}$$

and

$$U(f, P'') - L(f, P'') < \frac{\varepsilon}{2}.$$

Let $P = P' \cup P''$. This is a partition of $[a, b]$, and the upper and lower sums split at c:

$$U(f, P) = U(f, P') + U(f, P''),$$
$$L(f, P) = L(f, P') + L(f, P'').$$

It follows that

$$\begin{aligned} U(f, P) - L(f, P) &= [U(f, P') - L(f, P')] \\ &\quad + [U(f, P'') - L(f, P'')] \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

Hence f is Riemann integrable on $[a, b]$. □

Example 10. *Consider the function $f : [0, 1] \to \mathbb{R}$ defined by*

$$f(x) = \begin{cases} 0, & 0 \leq x < \frac{1}{2}, \\ 1, & \frac{1}{2} \leq x \leq 1. \end{cases}$$

This function is not continuous at $x = \frac{1}{2}$. Nevertheless, we will show that it is Riemann integrable.

Proof. Let $\varepsilon > 0$. Choose $0 < \delta < \frac{1}{2}$ such that $\delta < \varepsilon$, and consider the partition

$$P = \left\{0, \frac{1}{2} - \delta, \frac{1}{2}, 1\right\}.$$

On $[0, \frac{1}{2} - \delta]$, the function is constantly 0, so the supremum and infimum are both 0.

On $[\frac{1}{2} - \delta, \frac{1}{2}]$, the supremum is 1 and the infimum is 0.

Finally, on $[\frac{1}{2}, 1]$, the function is constantly 1, so the supremum and infimum are both 1.

Therefore,

$$\begin{aligned} U(f,P) - L(f,P) &= \sum_{i=1}^{3} (M_i - m_i)(t_i - t_{i-1}) \\ &= 0 + \delta + 0 \\ &= \delta \\ &< \varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ was arbitrary, f is Riemann integrable on $[0,1]$. □

11 Fundamental theorem of Calculus

Theorem 33 (Fundamental Theorem of Calculus)**.** *Let f be integrable on $[a,b]$, and define*

$$F(x) = \int_a^x f(t)\,dt.$$

If f is continuous at $c \in (a,b)$, then F is differentiable at c, and

$$F'(c) = f(c).$$

At the endpoints, the same conclusion holds for the appropriate one-sided derivatives.

Proof. Suppose first that $c \in (a,b)$ and let $h > 0$ be sufficiently small. By the additivity of the integral,

$$F(c+h) - F(c) = \int_c^{c+h} f(t)\,dt.$$

Let

$$m_h = \inf_{t \in [c, c+h]} f(t), \qquad M_h = \sup_{t \in [c, c+h]} f(t).$$

Since

$$m_h h \leq \int_c^{c+h} f(t)\,dt \leq M_h h,$$

we obtain

$$m_h \leq \frac{F(c+h) - F(c)}{h} \leq M_h.$$

Now suppose $h < 0$. Define m_h and M_h as the infimum and supremum of f on $[c+h, c]$. Then

$$m_h(-h) \leq \int_{c+h}^{c} f(t)\,dt \leq M_h(-h).$$

Using

$$F(c+h) - F(c) = -\int_{c+h}^{c} f(t)\,dt$$

and dividing by $h < 0$, we again obtain

$$m_h \leq \frac{F(c+h) - F(c)}{h} \leq M_h.$$

Since f is continuous at c, both m_h and M_h approach $f(c)$ as $h \to 0$. Therefore, by the Squeeze Theorem,

$$F'(c) = \lim_{h \to 0} \frac{F(c+h) - F(c)}{h} = f(c).$$

The proof at a and b is identical, using one-sided limits. □

Corollary 5. *If f is continuous on $[a, b]$, then*

$$F(x) = \int_{a}^{x} f(t)\,dt$$

is an antiderivative of f. In particular, if G is any antiderivative of f, then

$$\int_{a}^{b} f(t)\,dt = G(b) - G(a).$$

Proof. By the Fundamental Theorem of Calculus, $F'(x) = f(x)$ for every $x \in (a, b)$. Since $G'(x) = f(x)$ as well, we have

$$(G - F)'(x) = 0.$$

Thus, by the Mean Value Theorem, $G - F$ is constant on $[a, b]$. In particular,

$$G(b) - F(b) = G(a) - F(a).$$

Since $F(a) = 0$ and

$$F(b) = \int_{a}^{b} f(t)\,dt,$$

we conclude that

$$\boxed{\int_{a}^{b} f(t)\,dt = G(b) - G(a).}$$

□

Examples: Area vs integrals

1. $f(x) = x^3$. Its integral is $\frac{x^4}{4} + c$, however the area of the curve between -2, 2 is not $\frac{2^4}{4} - \frac{(-2)^4}{4} = 0$ as area always means something positive.

2. From fundamental theorem of Calculus, we can say that $\int x^n = \frac{x^{n+1}}{n+1}$.

Problem 65. *Find the area bounded by the curves $f(x) = x^3 - x$ and $g(x) = x^2$.*

Remark 7. *Let f be a bounded function on $[a, b]$. Even if f is not integrable, the quantities*

$$\sup_P L(f, P) \qquad \text{and} \qquad \inf_P U(f, P)$$

exist. They are called the lower integral *and* upper integral *of f, respectively, and are denoted by*

$$\mathrm{L}\int_a^b f \qquad \text{and} \qquad \mathrm{U}\int_a^b f.$$

These satisfy properties similar to those of the integral. In particular, if $a < c < b$, then

$$\mathrm{L}\int_a^b f = \mathrm{L}\int_a^c f + \mathrm{L}\int_c^b f,$$

and

$$\mathrm{U}\int_a^b f = \mathrm{U}\int_a^c f + \mathrm{U}\int_c^b f.$$

Moreover, if $m \le f(x) \le M$ for every $x \in [a, b]$, then

$$m(b-a) \le \mathrm{L}\int_a^b f \le \mathrm{U}\int_a^b f \le M(b-a).$$

Recall that f is integrable precisely when

$$\mathrm{L}\int_a^b f = \mathrm{U}\int_a^b f.$$

In this case, their common value is $\int_a^b f$.

12 Trignometric functions

We would finally like to use our understanding of Calculus so far in figuring what the functions $\sin x, \cos x, \tan x$ and so on are.

Before we begin, our intuition tells us is that the sin function is taken on an angle. So before we begin to understand that, we need to ask ourselves, what does an angle mean? How do we quantify our understanding of an angle?

12.1 What does angle mean?

Whatever angle means, we know that it should be between two lines and a point. Moreover two parallel curves given by purple and green below should measure the same quantity.

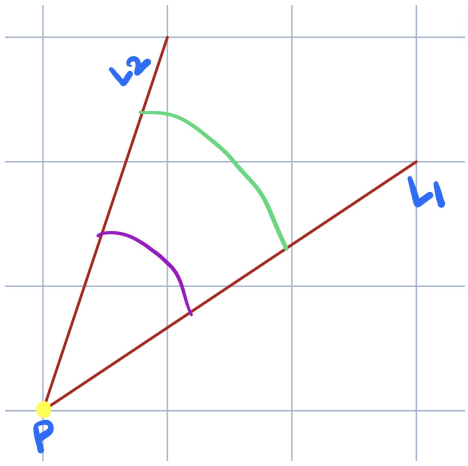


Figure 7: Angle

So, the above picture tells us that input data of an angle must be (P, L_1, L_2), that is a point P and two lines L_1, L_2. But if you see the figure below, we definitely know that something that the red curve quantifies should not be the same as what green curve quantifies, and the measurement of angle must be dependent on the green curve. Can you think of why the green one is better suited?

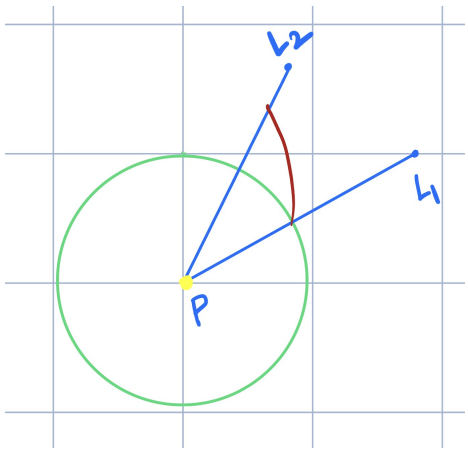


Figure 8: Angle 2

We use the standard fact from Euclidean geometry that a unique circle passes once you fix center and 1 point on the circle. This is what makes the data above precise.

Since whatever angle means should not distinguish between two parallel curves, we choose a circle of radius 1. We can also assume that our first line is the x-axis. Hence, the data of an angle is uniquely determined by three points of data (P, L_1, L_2). Now imagine L_1 is the x-axis, then the line L_2 will be determined by a point (x, y) in the unit circle satisfying $x^2 + y^2 = 1$. Whatever angle means, $\cos((P, L_1, L_2 = (x, y)))$ should be given by x and sin of this angle should be given by y.

With these remarks as background, we now begin the rigorous definition of the functions sin and cos. We first define π as the area of the unit circle, or equivalently, twice the area of a semicircle.

Definition 33. *We define*

$$\boxed{\pi = 2\int_{-1}^{1} \sqrt{1-x^2}\, dx.}$$

The definition of π above will allow us to define the trigonometric functions sin and cos rigorously, initially on the interval $[0, \pi]$.

We begin by defining a function $A(x)$ for $-1 \leq x \leq 1$. Geometrically, $A(x)$ represents the area of the sector of the unit circle bounded by the positive horizontal axis and the line joining the origin to the point $(x, \sqrt{1-x^2})$.

For $0 \leq x \leq 1$, this area can be obtained by adding the area of a right triangle to the area under the upper semicircle from x to 1. Thus,

$$A(x) = \frac{x\sqrt{1-x^2}}{2} + \int_x^1 \sqrt{1-t^2}\, dt.$$

The same expression remains valid for $-1 \leq x \leq 0$. Indeed, in this case the first term is negative, and its absolute value is precisely the area of the triangle that must be removed from the region under the semicircle.

Definition 34. *For $-1 \leq x \leq 1$, define*

$$\boxed{A(x) = \frac{x\sqrt{1-x^2}}{2} + \int_x^1 \sqrt{1-t^2}\, dt.}$$

Proposition 6. *The function A is continuous on $[-1, 1]$ and differentiable on $(-1, 1)$, with*

$$A'(x) = -\frac{1}{2\sqrt{1-x^2}}.$$

In particular, A is strictly decreasing on $[-1, 1]$, with $A(-1) = \pi/2$ and $A(1) = 0$.

Proof. Continuity follows from the definition, since both terms are continuous on $[-1, 1]$.

For $-1 < x < 1$, the product rule, the chain rule, and the Fundamental Theorem of Calculus give

$$\begin{aligned} A'(x) &= \frac{1}{2}\left(\sqrt{1-x^2} - \frac{x^2}{\sqrt{1-x^2}}\right) - \sqrt{1-x^2} \\ &= \frac{1-2x^2}{2\sqrt{1-x^2}} - \sqrt{1-x^2} \\ &= -\frac{1}{2\sqrt{1-x^2}}. \end{aligned}$$

Since $A'(x) < 0$ for every $x \in (-1, 1)$, the Mean Value Theorem implies that A is strictly decreasing.

Finally, evaluating at the endpoints gives

$$A(-1) = \int_{-1}^{1} \sqrt{1-t^2}\, dt = \frac{\pi}{2}, \qquad A(1) = 0.$$

□

Consequently, A maps $[-1, 1]$ continuously and strictly decreasingly onto $[0, \pi/2]$.

We now want to define $\cos x$ and $\sin x$ for $0 \le x \le \pi$ as the coordinates of the unique point $P = (\cos x, \sin x)$ on the upper half of the unit circle such that the sector between the positive horizontal axis and the line joining the origin to P has area $x/2$.

Definition 35. *For $0 \le x \le \pi$, define $\cos x$ to be the unique number in $[-1, 1]$ satisfying*

$$A(\cos x) = \frac{x}{2}.$$

We then define

$$\sin x = \sqrt{1 - (\cos x)^2}.$$

We should first verify that this definition makes sense. Recall that A is continuous and strictly decreasing on $[-1, 1]$, with

$$A(-1) = \frac{\pi}{2}, \qquad A(1) = 0.$$

For every $x \in [0, \pi]$, we have $x/2 \in [0, \pi/2]$. Thus, by the Intermediate Value Theorem, there exists some $y \in [-1, 1]$ such that

$$A(y) = \frac{x}{2}.$$

Since A is strictly decreasing, this y is unique. Hence $\cos x$, and consequently $\sin x$, are well-defined.

Theorem 34. *For every $x \in (0, \pi)$, the functions* sin *and* cos *are differentiable, and*

$$\boxed{\cos'(x) = -\sin x, \qquad \sin'(x) = \cos x.}$$

Proof. Let $B = 2A$. By definition,

$$B(\cos x) = x.$$

Thus, cos is the inverse function of B.

From our earlier computation,

$$A'(x) = -\frac{1}{2\sqrt{1-x^2}},$$

so

$$B'(x) = -\frac{1}{\sqrt{1-x^2}}, \qquad -1 < x < 1.$$

Since B' is nonzero on $(-1, 1)$, the inverse function theorem gives

$$\begin{aligned} \cos'(x) &= (B^{-1})'(x) \\ &= \frac{1}{B'(B^{-1}(x))} \\ &= -\sqrt{1-(\cos x)^2} \\ &= -\sin x. \end{aligned}$$

Next, using the definition

$$\sin x = \sqrt{1-(\cos x)^2},$$

and applying the chain rule, we obtain

$$\begin{aligned} \sin'(x) &= \frac{-2\cos x \cdot \cos'(x)}{2\sqrt{1-(\cos x)^2}} \\ &= \frac{\cos x \sin x}{\sin x} \\ &= \cos x. \end{aligned}$$

Here division by $\sin x$ is valid because $\sin x > 0$ for $0 < x < \pi$. □

We can now use the preceding results to understand the behavior of the sine and cosine functions on $[0, \pi]$.

Since

$$\cos'(x) = -\sin x < 0, \qquad 0 < x < \pi,$$

the function $\cos x$ is strictly decreasing on $[0, \pi]$. Moreover, $\cos 0 = 1$ and $\cos \pi = -1$. Thus, there is exactly one point in $[0, \pi]$ at which $\cos x = 0$.

We claim that this point is $\pi/2$. Recall that

$$A(\cos x) = \frac{x}{2}.$$

In particular,

$$A(0) = \frac{\pi}{2}.$$

If $\cos y = 0$, then

$$y = 2\int_0^1 \sqrt{1-t^2}\,dt.$$

Since $\sqrt{1-t^2}$ is an even function, we have

$$\int_{-1}^0 \sqrt{1-t^2}\,dt = \int_0^1 \sqrt{1-t^2}\,dt.$$

Consequently,

$$y = \int_{-1}^1 \sqrt{1-t^2}\,dt = \frac{\pi}{2}.$$

Next, consider the derivative of $\sin x$. We have

$$\sin'(x) = \cos x,$$

and hence

$$\sin'(x)\begin{cases} > 0, & 0 < x < \pi/2, \\ < 0, & \pi/2 < x < \pi. \end{cases}$$

Therefore, $\sin x$ is strictly increasing on $[0, \pi/2]$ and strictly decreasing on $[\pi/2, \pi]$. Since

$$\sin 0 = 0, \qquad \sin\frac{\pi}{2} = 1, \qquad \sin\pi = 0,$$

the sine function increases from 0 to 1 and then decreases back to 0.

Extending sine and cosine beyond $[0, \pi]$.

We extend the definitions of sine and cosine to all real numbers in two stages.

Step 1: Extension to $[0, 2\pi]$.

For $\pi \le x \le 2\pi$, define

$$\boxed{\begin{aligned} \sin x &= -\sin(2\pi - x), \\ \cos x &= \cos(2\pi - x). \end{aligned}}$$

Together with the original definitions on $[0, \pi]$, this defines both functions on $[0, 2\pi]$.

Step 2: Extension to $\mathbb{R}$.

Given any $x \in \mathbb{R}$, choose an integer k such that

$$x = 2\pi k + x', \qquad 0 \le x' < 2\pi.$$

We then define

$$\boxed{\begin{aligned} \sin x &= \sin x', \\ \cos x &= \cos x'. \end{aligned}}$$

Thus, both sine and cosine are periodic functions with period 2π, meaning that

$$\sin(x + 2\pi) = \sin x, \qquad \cos(x + 2\pi) = \cos x$$

for every real number x.

These extensions preserve the basic properties of sine and cosine. In particular, the identity

$$\boxed{\sin^2 x + \cos^2 x = 1}$$

continues to hold for every $x \in \mathbb{R}$.

Furthermore, we have

$$\boxed{\sin'(x) = \cos x, \qquad \cos'(x) = -\sin x}$$

for all x that are not integer multiples of π.

To illustrate this, suppose $\pi < x < 2\pi$. By definition,

$$\sin x = -\sin(2\pi - x).$$

Applying the chain rule gives

$$\begin{aligned} \sin'(x) &= -\cos(2\pi - x) \cdot (-1) \\ &= \cos(2\pi - x) \\ &= \cos x. \end{aligned}$$

Similarly,

$$\begin{aligned} \cos'(x) &= -\sin(2\pi - x) \cdot (-1) \\ &= \sin(2\pi - x) \\ &= -\sin x. \end{aligned}$$

The same reasoning, together with periodicity, establishes these derivative formulas on the remaining open intervals between consecutive multiples of π.

13 Taylor series

Taylor series of a function and its existence has many different applications. We will use it to define the function e^x, which would also define the number e.

Let us first discuss approximating smooth functions upto n-th order.

Let us first discuss the case of polynomials. Given a polynomial

$$f(x) = a_0 + a_1 x + \cdots + a_n x^n,$$

we can choose any point c and rewrite it in the form

$$f(x) = b_0 + b_1(x - c) + b_2(x - c)^2 + \cdots + b_n(x - c)^n.$$

Why is this possible? Notice that $x = (x - c) + c$, so substituting this into the original expression and expanding using the binomial theorem gives a polynomial in powers of $(x - c)$.

How can we determine the coefficients b_i? Substituting $x = c$, we get

$$f(c) = b_0.$$

Differentiating once, we obtain

$$f'(x) = b_1 + 2b_2(x - c) + \cdots + nb_n(x - c)^{n-1}.$$

Setting $x = c$ gives $f'(c) = b_1$. Similarly, differentiating twice and evaluating at $x = c$, we get

$$f''(c) = 2b_2.$$

Continuing this process, we find that

$$f^{(k)}(c) = k!b_k.$$

Therefore,

$$\boxed{b_k = \frac{f^{(k)}(c)}{k!}, \qquad 0 \leq k \leq n.}$$

Consequently, every polynomial of degree at most n can be written as

$$\boxed{f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(c)}{k!}(x - c)^k.}$$

This is called the *Taylor expansion* of f about the point c.

13.0.1 Taylor Polynomials and Taylor's Theorem

Definition 36. *We say that a function $f : [a, b] \to \mathbb{R}$ is* smooth *if all iterated derivatives of f exist on (a, b).*

Definition 37. *Let $f : [a, b] \to \mathbb{R}$ be a smooth function and let $c \in (a, b)$. The* nth order Taylor polynomial *of f about c is defined by*

$$P_n f(x) = f(c) + f'(c)(x - c) + \frac{f''(c)}{2!}(x - c)^2 + \cdots + \frac{f^{(n)}(c)}{n!}(x - c)^n.$$

Equivalently, in summation notation,

$$\boxed{P_n f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(c)}{k!}(x - c)^k.}$$

Here, we use the convention $f^{(0)} = f$.

Theorem 35 (Taylor's Theorem)**.** *Let $f : [a, b] \to \mathbb{R}$ be a smooth function, and let $P_n f$ be its nth order Taylor polynomial about some $c \in (a, b)$. Then, for any $d \in (a, b)$ with $d \neq c$, there exists a point z strictly between c and d such that*

$$f(d) - P_n f(d) = \frac{f^{(n+1)}(z)}{(n+1)!}(d-c)^{n+1}.$$

Proof. For simplicity, write $P(x) = P_n f(x)$. By the definition of the Taylor polynomial, we know that

$$P^{(k)}(c) = f^{(k)}(c), \qquad k = 0, 1, \ldots, n.$$

Set

$$M = \frac{f(d) - P(d)}{(d-c)^{n+1}}.$$

Our goal is to show that

$$M = \frac{f^{(n+1)}(z)}{(n+1)!}$$

for some z between c and d.

Consider the auxiliary function

$$g(x) = f(x) - P(x) - M(x-c)^{n+1}.$$

By our choice of M, we have $g(d) = 0$. Moreover, for every $k = 0, 1, \ldots, n$,

$$g^{(k)}(c) = f^{(k)}(c) - P^{(k)}(c) = 0.$$

In particular, $g(c) = g(d) = 0$.

Applying Rolle's theorem to g on the interval with endpoints c and d, we obtain a point z_1 strictly between them such that $g'(z_1) = 0$. Since $g'(c) = 0$, another application of Rolle's theorem gives a point z_2 between c and z_1 such that $g''(z_2) = 0$.

Continuing in this way, and using $g^{(k)}(c) = 0$ at each step, we eventually obtain a point z strictly between c and d satisfying

$$g^{(n+1)}(z) = 0.$$

On the other hand, differentiating the expression for g exactly $n+1$ times gives

$$g^{(n+1)}(x) = f^{(n+1)}(x) - (n+1)!M,$$

since P has degree at most n.

Evaluating at z, we conclude that

$$0 = f^{(n+1)}(z) - (n+1)!M,$$

and hence

$$M = \frac{f^{(n+1)}(z)}{(n+1)!}.$$

Substituting this into the definition of M proves

$$f(d) - P_n f(d) = \frac{f^{(n+1)}(z)}{(n+1)!}(d-c)^{n+1}.$$

□

Definition 38. *The expression*

$$R_n(d) = \frac{f^{(n+1)}(z)}{(n+1)!}(d-c)^{n+1},$$

where z is some point between c and d, is called the Lagrange remainder *or* Lagrange error term *for the nth order Taylor approximation of f about c.*

The natural question to now ask if whether a function $f(x)$ can be expressed as some sort of an infinite sum via this process? This definitely means we have to test convergence and deal with infinite series and sequences of functions. First let us discuss sequences of functions and then come to series.

Definition 39. *A* sequence of functions $\{f_n\}$ *on $[a,b]$ is an infinite collection of functions*

$$f_1, f_2, f_3, \ldots,$$

where each $f_n : [a,b] \to \mathbb{R}$.

Given a sequence of functions $\{f_n\}$, fixing any point $x_0 \in [a,b]$ gives a sequence of real numbers

$$f_1(x_0), f_2(x_0), f_3(x_0), \ldots.$$

Thus, we can study the convergence of a sequence of functions by first considering the convergence of these numerical sequences at individual points.

Definition 40. *Let $\{f_n\}$ be a sequence of functions $f_n : [a,b] \to \mathbb{R}$.*

1. *We say that* $\{f_n\}$ converges pointwise *to a function $f : [a,b] \to \mathbb{R}$ if, for every $x_0 \in [a,b]$, the sequence of real numbers $\{f_n(x_0)\}$ converges to $f(x_0)$.*

   Equivalently, for every $x_0 \in [a,b]$ and every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that

   $$|f_n(x_0) - f(x_0)| < \epsilon \qquad \text{for all } n \geq N.$$

2. *We say that $\{f_n\}$ is* pointwise Cauchy *if, for every $x_0 \in [a,b]$, the sequence $\{f_n(x_0)\}$ is a Cauchy sequence of real numbers.*

   Equivalently, for every $x_0 \in [a,b]$ and every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that

   $$|f_n(x_0) - f_m(x_0)| < \epsilon \qquad \text{for all } m, n \geq N.$$

13.0.2 Pointwise Convergence Does Not Preserve Continuity

At first, pointwise convergence might seem like a natural way to define convergence for sequences of functions. However, it has some limitations, as the following example illustrates.

Example 11. *Consider the sequence of functions*

$$f_n : [0,1] \to \mathbb{R}, \qquad f_n(x) = x^n.$$

Notice that every f_n is continuous on $[0,1]$.

Let us determine the pointwise limit of this sequence. If $0 \leq x < 1$, then

$$\lim_{n\to\infty} x^n = 0.$$

On the other hand, when $x = 1$, we have

$$f_n(1) = 1$$

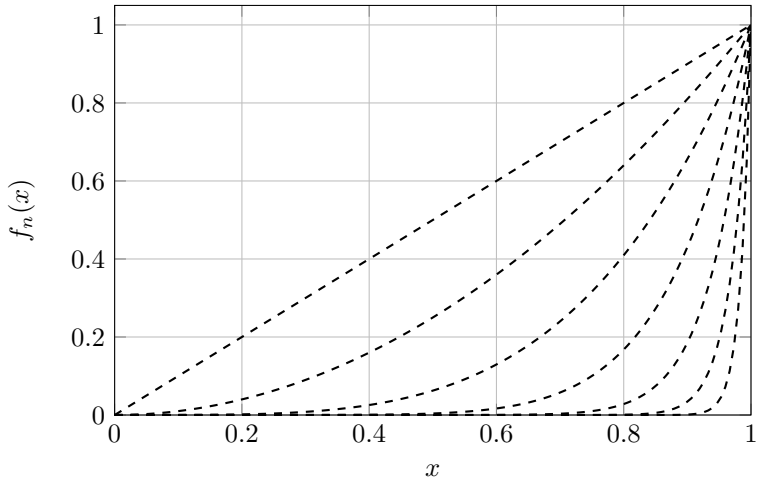
for every $n \in \mathbb{N}$. Therefore, the sequence $\{f_n\}$ converges pointwise to the function

$$f(x) = \begin{cases} 0, & 0 \leq x < 1, \\ 1, & x = 1. \end{cases}$$

However, the limiting function f is not continuous at $x = 1$, since

$$\lim_{x\to 1^-} f(x) = 0 \neq 1 = f(1).$$

Thus, even though every function in our sequence is continuous, its pointwise limit need not be continuous. In particular, this example also shows that a pointwise Cauchy sequence of continuous functions need not have a continuous limit.



Discussion. What went wrong in this example? At every fixed point $x \in [0, 1)$, the values x^n approach zero, but the functions always take the value 1 at $x = 1$.

This suggests that pointwise convergence may not give us enough control over how the functions behave across the entire interval. Can we introduce a stronger notion of convergence that preserves continuity?

Definition 41 (Uniform Convergence)**.** *Let $\{f_n\}$ be a sequence of functions $f_n : [a, b] \to \mathbb{R}$.*

1. *We say that $\{f_n\}$* converges uniformly *to a function $f : [a, b] \to \mathbb{R}$ if, for every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that*
$$|f_n(x) - f(x)| < \epsilon$$
for all $n \geq N$ and all $x \in [a, b]$.

2. *We say that $\{f_n\}$ is* uniformly Cauchy *if, for every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that*
$$|f_m(x) - f_n(x)| < \epsilon$$
for all $m, n \geq N$ and all $x \in [a, b]$.

Problem 66. *Show that if $f_n(x) = x^n$, and $f(x) = \lim_{n\to\infty} f_n(x)$, then $f_n(x)$ does not tend to $f(x)$ uniformly.*

Problem 67 (Continuous Limit Theorem)**.** *Let $\{f_n\}$ be a sequence of continuous functions $f_n : [a, b] \to \mathbb{R}$, and suppose that $\{f_n\}$ converges uniformly to a function $f : [a, b] \to \mathbb{R}$.*

Prove that f is continuous on $[a, b]$.

Before we start discussing convergence of *series* of functions, let us see an example, e^x!

Let us now try to construct a function f satisfying
$$f' = f.$$

Observation 1. Suppose such a function exists. Since $f' = f$, we have
$$f'' = f' = f,$$
and, continuing in this way,
$$f^{(n)} = f$$
for every $n \geq 1$. Thus, f must be smooth, and all its derivatives agree with the original function.

Observation 2. We claim that any such function must agree with its Taylor series.

Suppose f is defined on an interval $[a, b]$, and let $c \in (a, b)$. By Taylor's theorem, for any $x \in [a, b]$, the error in approximating $f(x)$ by its nth order Taylor polynomial is
$$f(x) - P_n f(x) = \frac{f^{(n+1)}(d)}{(n+1)!}(x - c)^{n+1}$$

for some point d between c and x.

Since every derivative of f equals f, this becomes

$$f(x) - P_n f(x) = \frac{f(d)}{(n+1)!}(x-c)^{n+1}.$$

Now, f is continuous on the closed interval $[a,b]$, so it is bounded. Hence, there exists $M > 0$ such that

$$|f(x)| \leq M \qquad \text{for all } x \in [a,b].$$

Consequently,

$$|f(x) - P_n f(x)| \leq \frac{M}{(n+1)!}|x-c|^{n+1}.$$

Notice that $|x-c| \leq b-a$ for every $x \in [a,b]$. Therefore,

$$|f(x) - P_n f(x)| \leq \frac{M(b-a)^{n+1}}{(n+1)!}.$$

As $n \to \infty$, the right-hand side approaches zero, because factorials grow faster than any fixed exponential sequence. In particular, the bound does not depend on x.

Thus, the Taylor polynomials $P_n f$ converge uniformly to f on $[a,b]$.

Since $f^{(n)}(c) = f(c)$ for every n, its Taylor polynomial takes the form

$$P_n f(x) = f(c)\left(1 + (x-c) + \frac{(x-c)^2}{2!} + \cdots + \frac{(x-c)^n}{n!}\right).$$

Passing to the limit, we obtain

$$f(x) = f(c)\left(1 + (x-c) + \frac{(x-c)^2}{2!} + \frac{(x-c)^3}{3!} + \cdots\right).$$

In particular, taking $c = 0$ and assuming $f(0) = 1$, we are led to the series

$$\boxed{\exp(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}.}$$

We can define this series for every $x \in \mathbb{R}$. Moreover, on any bounded interval $[-R,R]$, we have

$$\left|\frac{x^n}{n!}\right| \leq \frac{R^n}{n!}.$$

Since the numerical series

$$\sum_{n=0}^{\infty} \frac{R^n}{n!}$$

converges, the series defining $\exp(x)$ converges uniformly on every bounded interval.

The series of derivatives,

$$\sum_{n=1}^{\infty} \frac{x^{n-1}}{(n-1)!},$$

also converges uniformly on every bounded interval. Using the theorem on differentiating uniformly convergent series, we obtain

$$\frac{d}{dx}\exp(x) = \sum_{n=1}^{\infty} \frac{x^{n-1}}{(n-1)!} = \sum_{n=0}^{\infty} \frac{x^n}{n!} = \exp(x).$$

Thus, we have constructed a smooth function on $\mathbb{R}$ whose derivative is itself. Furthermore, our earlier argument shows that every solution of $f' = f$ on an interval containing 0 must be of the form

$$f(x) = f(0)\exp(x).$$

Observation 3. We now claim that the function constructed above satisfies the familiar identity

$$\boxed{\exp(x+y) = \exp(x)\exp(y)}$$

for every $x, y \in \mathbb{R}$.

Proof. Recall that

$$\exp(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

For any $x, y \in \mathbb{R}$, we have

$$\exp(x+y) = \sum_{n=0}^{\infty} \frac{(x+y)^n}{n!}.$$

By the binomial theorem,

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

Therefore,

$$\begin{aligned}
\exp(x+y) &= \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k} \\
&= \sum_{n=0}^{\infty} \sum_{k=0}^{n} \frac{x^k}{k!} \frac{y^{n-k}}{(n-k)!} \\
&= \left(\sum_{k=0}^{\infty} \frac{x^k}{k!} \right) \left(\sum_{j=0}^{\infty} \frac{y^j}{j!} \right) \\
&= \exp(x)\exp(y).
\end{aligned}$$

The rearrangement of the sums is justified because both exponential series converge absolutely.

Hence,

$$\boxed{\exp(x+y) = \exp(x)\exp(y).}$$

□

Problem 68. *Suppose $g : \mathbb{R} \to \mathbb{R}$ is a continuous function satisfying*

$$g(x+y) = g(x)g(y)$$

for all $x, y \in \mathbb{R}$.

Show that either g is identically zero, or

$$g(x) = g(1)^x$$

for every $x \in \mathbb{R}$.

Hint: First, show that if g is not identically zero, then $g(0) = 1$ and $g(x) > 0$ for every $x \in \mathbb{R}$. Next, prove that

$$g(n) = g(1)^n$$

for every natural number n. Then establish that

$$g\left(\frac{1}{n}\right) = g(1)^{1/n}.$$

Use these observations to show that $g(r) = g(1)^r$ for every rational number r. Finally, use the density of $\mathbb{Q}$ in $\mathbb{R}$ and continuity to extend the identity to all real numbers.

We can now apply this result to the function

$$\exp(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots .$$

Recall that we have already established that exp is continuous and satisfies

$$\exp(x+y) = \exp(x)\exp(y).$$

Moreover, $\exp(0) = 1$, so this function is not identically zero. By the previous problem, we must therefore have

$$\exp(x) = \exp(1)^x.$$

Notice that

$$\exp(1) = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{1}{n!}.$$

We give this number a special name:

$$\boxed{e = \exp(1) = \sum_{n=0}^{\infty} \frac{1}{n!}.}$$

Consequently, the function we constructed using the power series is precisely

$$\boxed{\exp(x) = e^x.}$$

Recall that our original goal was to find a function whose derivative equals itself. Since we have already proved that $\exp'(x) = \exp(x)$, we now obtain

$$\boxed{\frac{d}{dx}e^x = e^x.}$$

Thus, starting with the simple question of finding a function equal to its own derivative, we have arrived at the exponential function and the number e.

This also gives us a way to understand why the exponential function has such a special role in calculus: it is the unique solution of $f' = f$ satisfying $f(0) = 1$.

Once we have established a definition of e, let us prove that it is irrational!

13.1 Problem Set 3

Problem 69. *Consider the function $f : [0, 1] \to \mathbb{R}$ defined by*

$$f(x) = \begin{cases} x + \lfloor x \rfloor, & \text{if } x \in \mathbb{Q}, \\ 0, & \text{if } x \notin \mathbb{Q}, \end{cases}$$

where $\lfloor x \rfloor$ denotes the greatest integer less than or equal to x.

Prove that f is not Riemann integrable on $[0, 1]$ by showing that the difference between its upper and lower sums cannot be made arbitrarily small, regardless of the choice of partition.

Problem 70. *1. Let f be a smooth function such that for every value of x and for every n, $|f^{(n)}(x)| \leq M$ (this means independent of the choice of n, x.) Prove that $\lim_{n\to\infty} P_n f(x) = f(x)$. [Hint: Use Taylor's theorem!]*

2. Does a Taylor series of $\sin x$ and $\sin(x^2)$ exist?

We cannot always compute integrals exactly, but we can do an approximation!

Problem 71. *Let $f(t) = \int_0^t \sin(x^2)$. Is f smooth? What will its power series look like? Every single step must be rigorously justified.*

Compute the integral $\int_0^1 \sin(x^2)$ upto an error factor of $\frac{1}{8!}$.

Problem 72. *Consider the function*

$$f : [-\pi/2, \pi/2] \longrightarrow [-1, 1], \qquad f(x) = \sin x.$$

1. *Explain why f has an inverse function. Would an inverse still exist if we considered* $\sin : \mathbb{R} \to [-1, 1]$ *instead? Why or why not?*

2. *We denote the inverse of f by*

$$\arcsin : [-1, 1] \longrightarrow [-\pi/2, \pi/2].$$

   Show that arcsin *is differentiable on $(-1, 1)$ and compute its derivative.*

3. *Suppose we extend the domain of f to $[-1, 2]$. Does the function*

$$f : [-1, 2] \longrightarrow [-1, 1], \qquad f(x) = \sin x$$

   still have an inverse? Justify your answer.

4. *Compute the integral $\int_0^1 \dfrac{1}{\sqrt{1-x^2}}$ using the technique of u-substitution discussed in section.*

•

Problem 73 (Integration by Parts)**.** *Let $f, g : [a, b] \to \mathbb{R}$ be continuous functions, and suppose that F and G are antiderivatives of f and g, respectively. Prove that for every $x \in [a, b]$,*

$$\int_a^x f(t)G(t)\,dt = F(x)G(x) - F(a)G(a) - \int_a^x F(t)g(t)\,dt.$$

Hint: Recall the product rule for differentiation. Apply it to $F(t)G(t)$, and then use the Fundamental Theorem of Calculus.

Problem 74. *Let $p > 0$ be a real number. Prove rigorously that the series*

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

converges if and only if $p > 1$, and diverges otherwise, by comparing it with the Riemann sums for the function

$$f(x) = \frac{1}{x^p}.$$

Problem 75. *Let $0 < a < b$. Use integration by parts to show that*

$$\int_a^b \frac{\sin x}{x}\,dx = \frac{\cos a}{a} - \frac{\cos b}{b} - \int_a^b \frac{\cos x}{x^2}\,dx.$$

Using this identity, prove that the improper integral

$$\int_0^\infty \frac{\sin x}{x}\, dx$$

converges.

Hint: First consider the limit as $a \to 0^+$. Why does the integrand on the left-hand side not cause any problems near 0? Next, use the expression on the right-hand side to study the limit as $b \to \infty$.

References

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[Spi06] Spivak, Michael. *Calculus*. Cambridge University Press, 2006.